Linear wave theory





MEC_5EO04_TA/MEC_53456_EP – Class 2
October 9, 2026

Marissa L. Yates

marissa.yates@enpc.fr

Course schedule

I. Wave dynamics	Project	Lecturer	Date
1. Introduction and characterizing ocean waves and sea states	Description of class project and site selection	M. Yates	25/09
2. Linear wave theory	Characterizing offshore waves	M. Yates	09/10
3. Nearshore wave propagation	Calculating extreme wave conditions	M. Yates	17/10
4. Coastal hydrodynamics	**Student mini-presentations**	M. Yates	06/11**
II. Numerical wave modeling			
5. Numerical wave modeling I	Introduction to wave propagation model	J. Harris	23/10**
6. Numerical wave modeling II	Estimating mean and extreme nearshore waves	J. Harris	13/11
III. Wave-structure interactions			
7. Dynamics of a body in waves	Wave load estimation	J. Harris	20/11
8. Modeling wave-body interactions	Calculating movement of floating wind turbine	C. Peyrard	04/12
9. Offshore wind turbine foundations	Final project work session	C. Peyrard	11/12
10. Final project presentations 14h - 18h15			***18/12***

Objectives

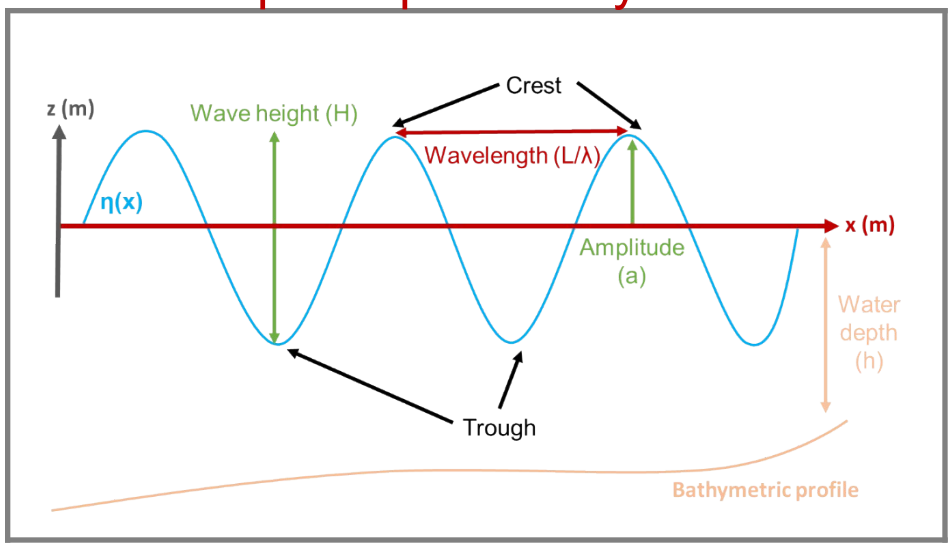
- Describe the spatial and temporal evolution of waves using linear wave theory
- Derive the dispersion relation
- Be able to describe wave kinematics and wave pressure fluctuations
- Apply wave energy conservation and evaluate the impacts on wave processes



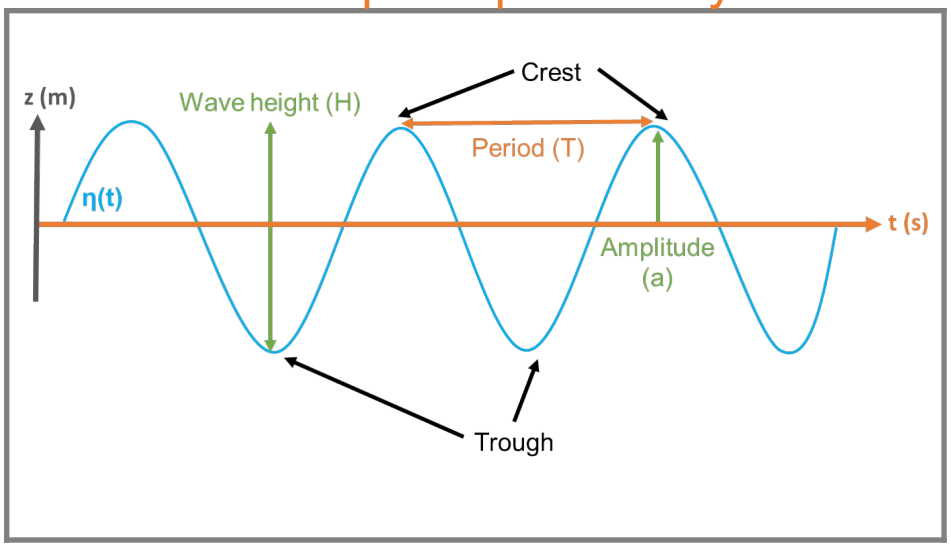


Review : spatial and temporal periodicity of waves

Spatial periodicity



Temporal periodicity



Wavelength = L or λ (m)
Wave number = k = 2π/L (rad/m)

Wave period= T (s)
Frequency = f = 1/T (Hz)
Frequency = ω = 2π/T (rad/s)

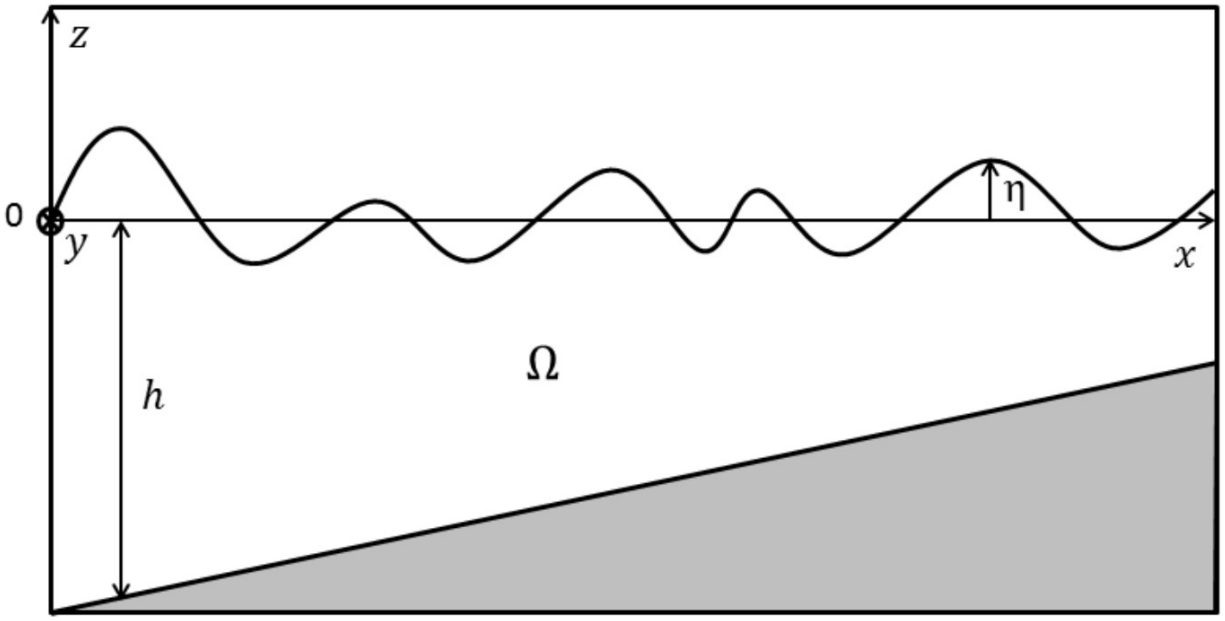
Link between the spatial and temporal domains:

Dispersion relation

$$\omega^2 = gk\tanh(kh)$$

Linear wave theory

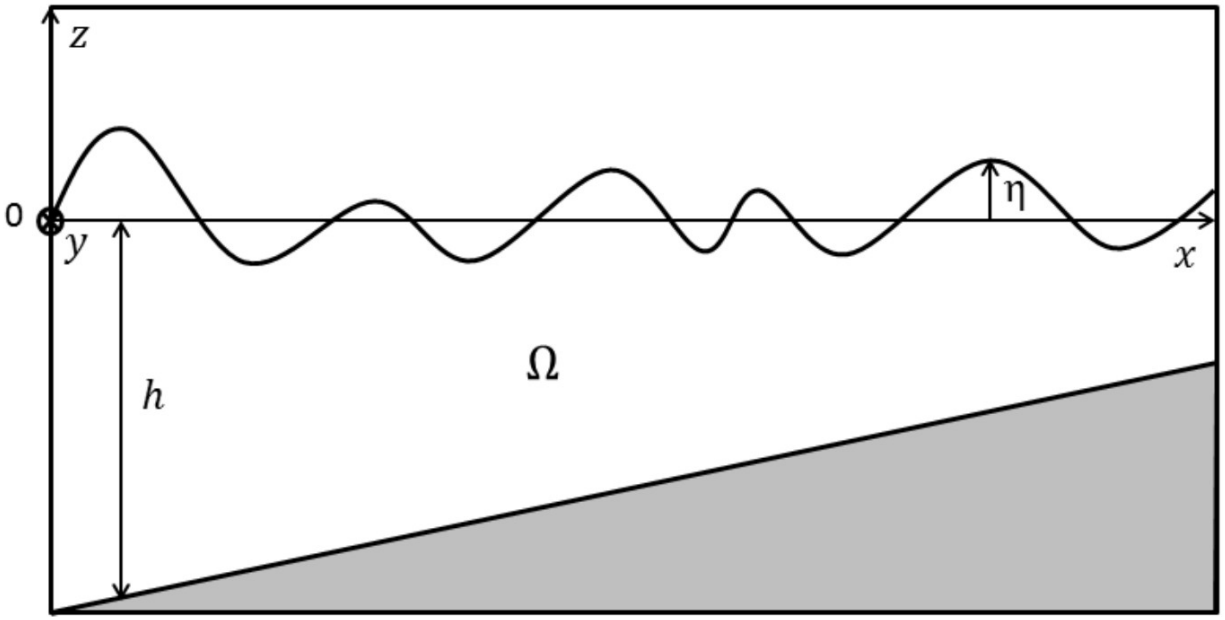
Navier Stokes to Euler equations



In Ω:

- Conservation of mass (continuity equation)
- Conservation of momentum

Navier Stokes to Euler equations

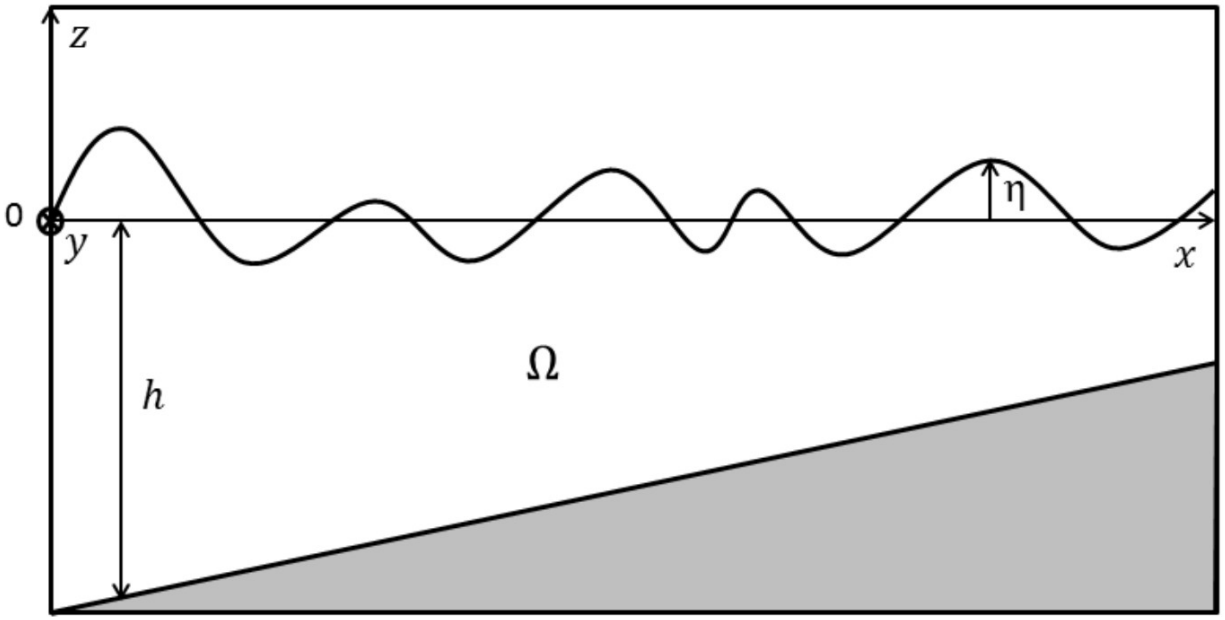


H1: Incompressible flow

$$\frac{\partial \rho}{\partial t} + \nabla.(\rho\, \underline{\mathbf{v}}) = 0 \qquad \nabla \equiv (\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z})^T$$

$$\frac{\partial \rho \underline{\mathbf{v}}}{\partial t} + \nabla.(\rho \underline{\mathbf{v}} \otimes \underline{\mathbf{v}}) = -\nabla p + \nabla.\underline{\underline{\tau}} + \rho \underline{g}$$

Navier Stokes to Euler equations

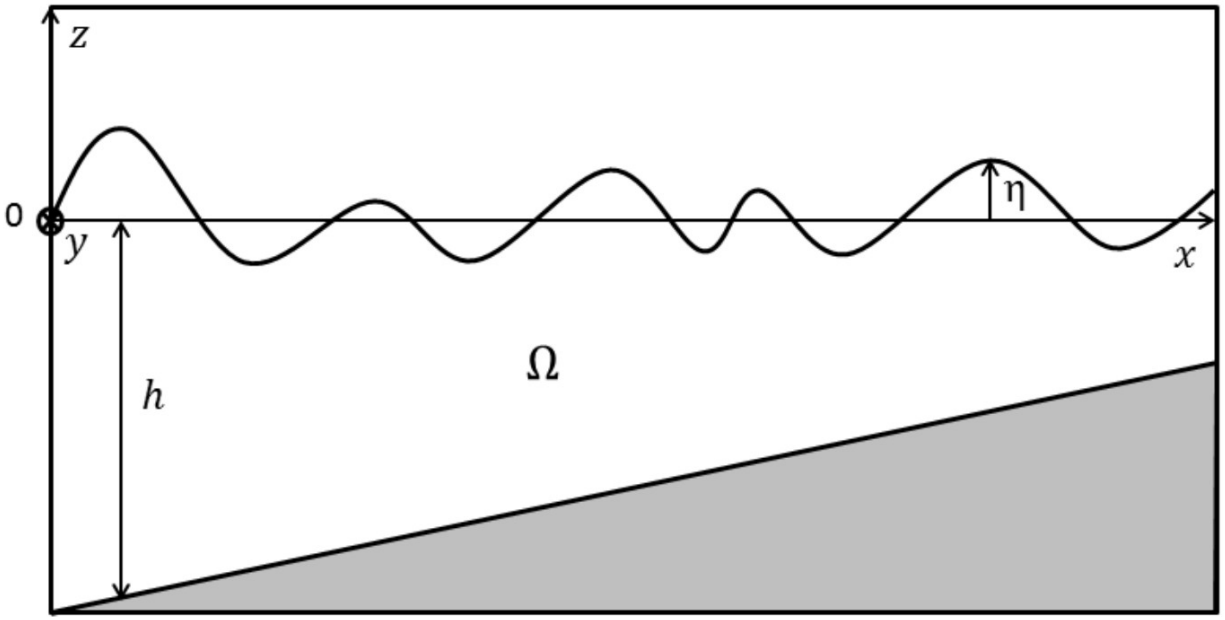


$$\nabla . \underline{\mathbf{v}} = 0$$

H2: Inviscid fluid

$$\frac{\partial \underline{\mathbf{v}}}{\partial t} + (\underline{\mathbf{v}}.\nabla)\underline{\mathbf{v}} = -\frac{1}{\rho}\nabla p + \nu \Delta \underline{\mathbf{v}} + \underline{g}$$

Navier Stokes to Euler equations



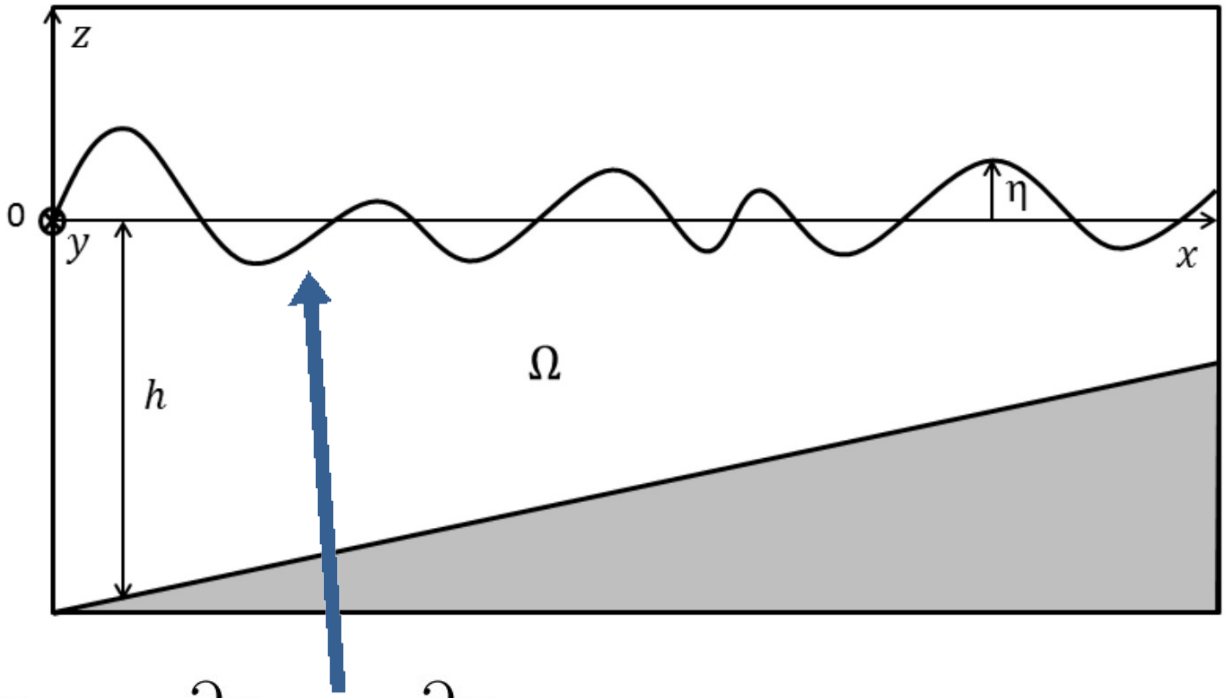
In Ω:

$$\nabla . \underline{\mathbf{v}} = 0$$

$$\frac{\partial \underline{\mathbf{v}}}{\partial t} + \underline{\mathbf{v}} \cdot \nabla \underline{\mathbf{v}} = -\frac{1}{\rho} \nabla p + \underline{g}$$

Euler equations

Boundary conditions: free surface

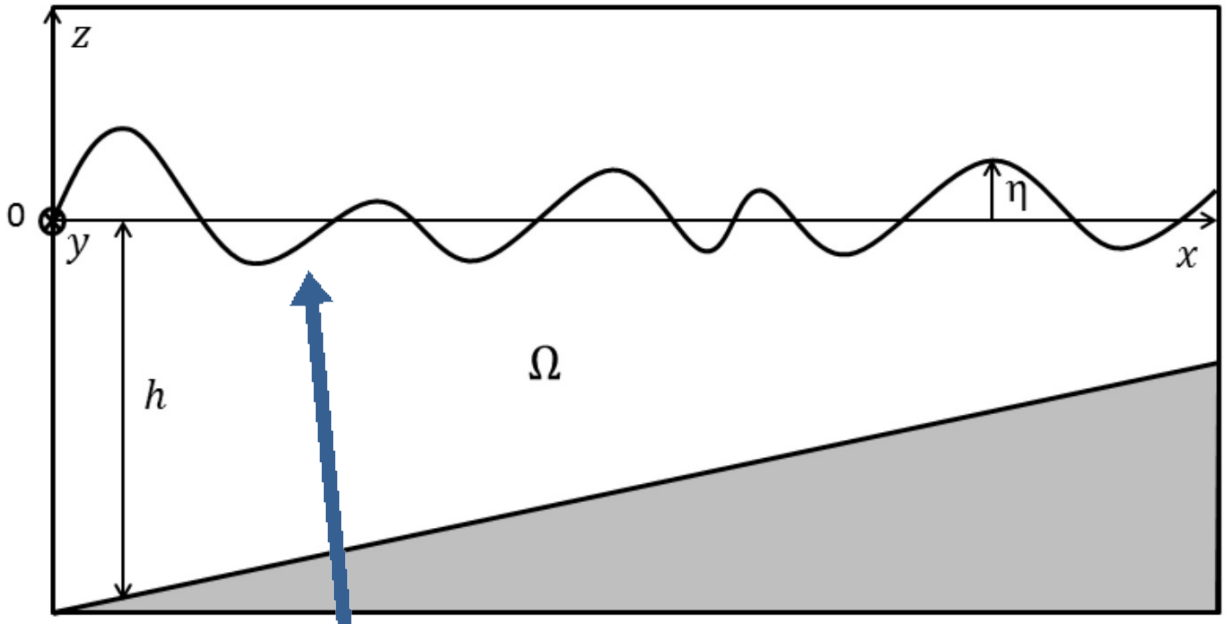


$$-\frac{\partial \eta}{\partial t} - u\frac{\partial \eta}{\partial x} - v\frac{\partial \eta}{\partial y} + w = 0 \qquad \text{at } z = \eta(x, y, t)$$

$$-\frac{\partial \eta}{\partial t} - \underline{\mathrm{v}}_{\mathrm{H}}.\nabla_{\mathrm{H}}\eta + w = 0$$

Kinematic (KFSBC)

Boundary conditions: free surface

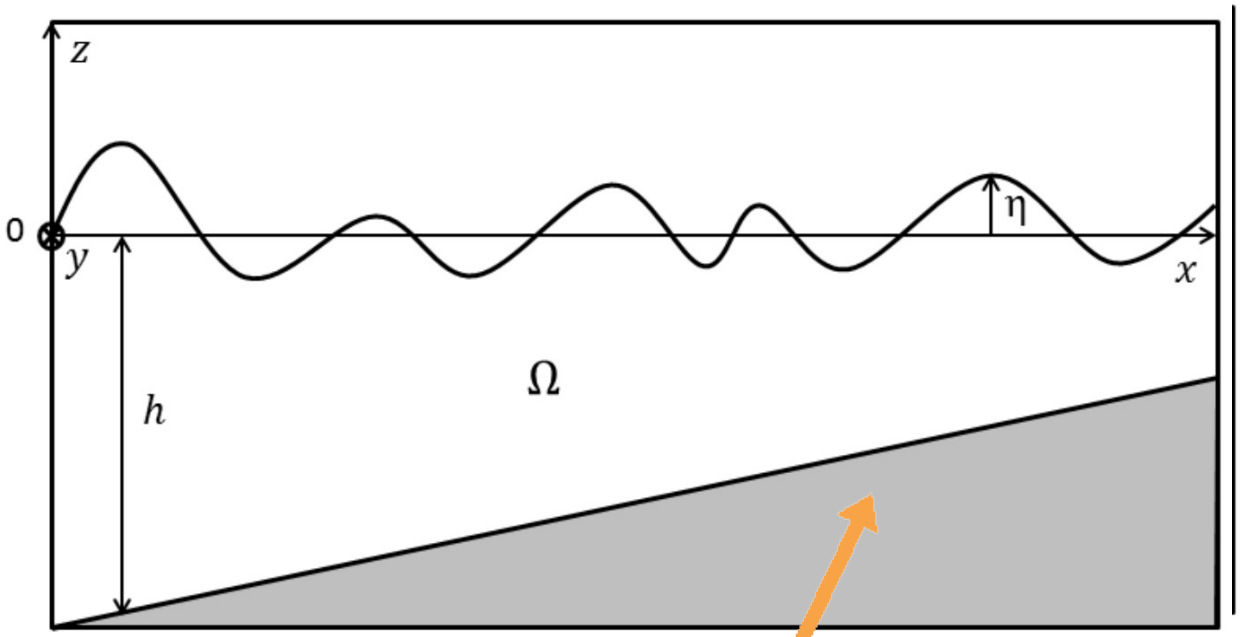


$$p(x, y, \eta(x, y, t), t) = p_{atm}(x, y, t)$$

$$z = \eta(x, y, t)$$

Dynamic (DFSBC)

Boundary conditions: bottom

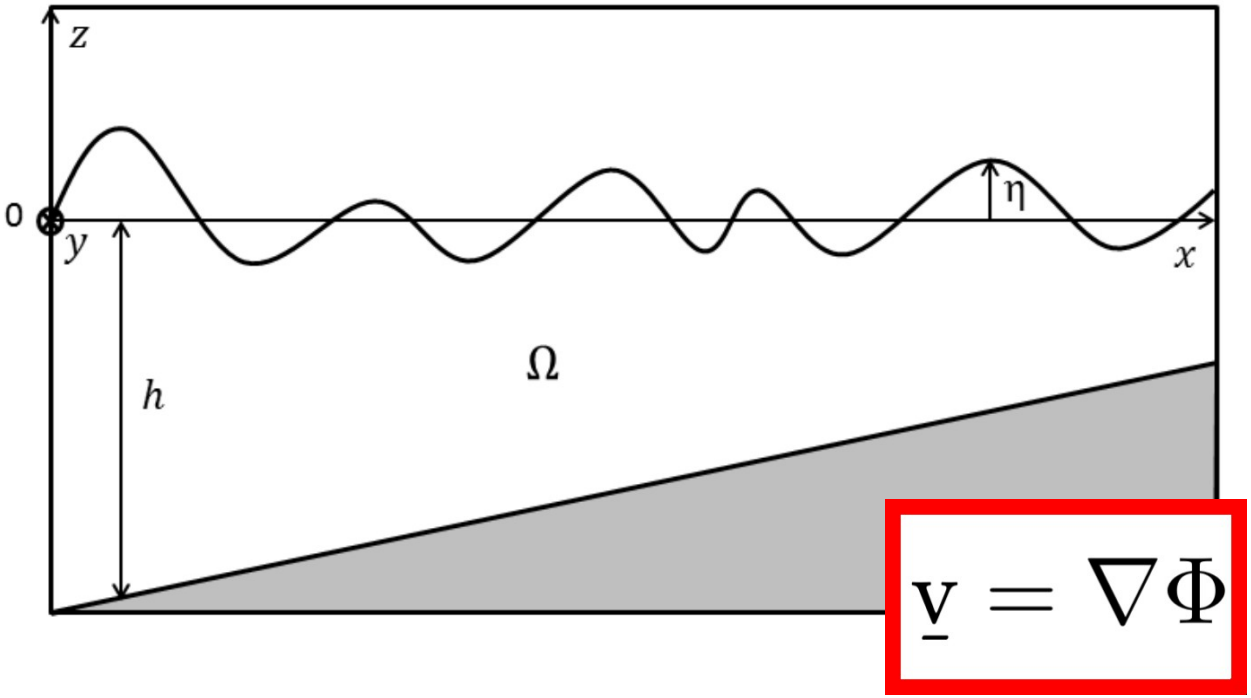


$$\frac{\partial h}{\partial t} + u\frac{\partial h}{\partial x} + v\frac{\partial h}{\partial y} + w = 0 \qquad z = -h(x, y, t)$$

$$\frac{\partial h}{\partial t} + \underline{\mathbf{v}}_{\mathrm{H}} . \nabla_{\mathrm{H}} h + w = 0$$

Kinematic (BBC)

Navier Stokes to Euler equations



$$\underline{v} = \nabla\Phi$$

H3: Irrotational flow

$$\nabla . \underline{v} = 0$$

$$\frac{\partial \underline{v}}{\partial t} + (\underline{v}.\nabla)\underline{v} = -\frac{1}{\rho}\nabla p + \underline{g}$$

Euler equations

Potential flow: water wave equations

$$\frac{\partial \underline{v}}{\partial t} + (\underline{v}.\nabla)\underline{v} = -\frac{1}{\rho}\nabla p + \underline{g} \qquad \boxed{\underline{v} = \nabla\Phi}$$

$$\Leftrightarrow \quad \frac{\partial \nabla\Phi}{\partial t} + \nabla\Phi.\nabla(\nabla\Phi) = -\frac{1}{\rho}\nabla p - \nabla(gz)$$

$$\Leftrightarrow \quad \nabla\left(\frac{\partial\Phi}{\partial t} + \frac{1}{2}(\nabla\Phi)^2 + \frac{p}{\rho} + gz\right) = 0$$

$$\boxed{\frac{\partial\Phi}{\partial t} + \frac{1}{2}(\nabla\Phi)^2 + \frac{p}{\rho} + gz = 0}$$

Nonlinear potential flow problem

$$\Delta\Phi = 0 \qquad \text{in } \Omega$$

$$\frac{\partial \eta}{\partial t} + \nabla_{\mathrm{H}}\Phi.\nabla_{\mathrm{H}}\eta - \frac{\partial \Phi}{\partial z} = 0$$

$$\frac{\partial \Phi}{\partial t} + \frac{1}{2}(\nabla\Phi)^2 + g\eta = -\frac{p_{atm}(x,y,t)}{\rho} \qquad \text{at } z = \eta(x,y,t)$$

$p_{atm} = 0$

$$\frac{\partial h}{\partial t} + \nabla_{\mathrm{H}}\Phi.\nabla_{\mathrm{H}}h + \frac{\partial \Phi}{\partial z} = 0 \qquad \text{at } z = -h(x,y,t)$$

Nonlinear potential flow problem

$$\Delta\Phi = 0 \qquad \text{in } \Omega$$

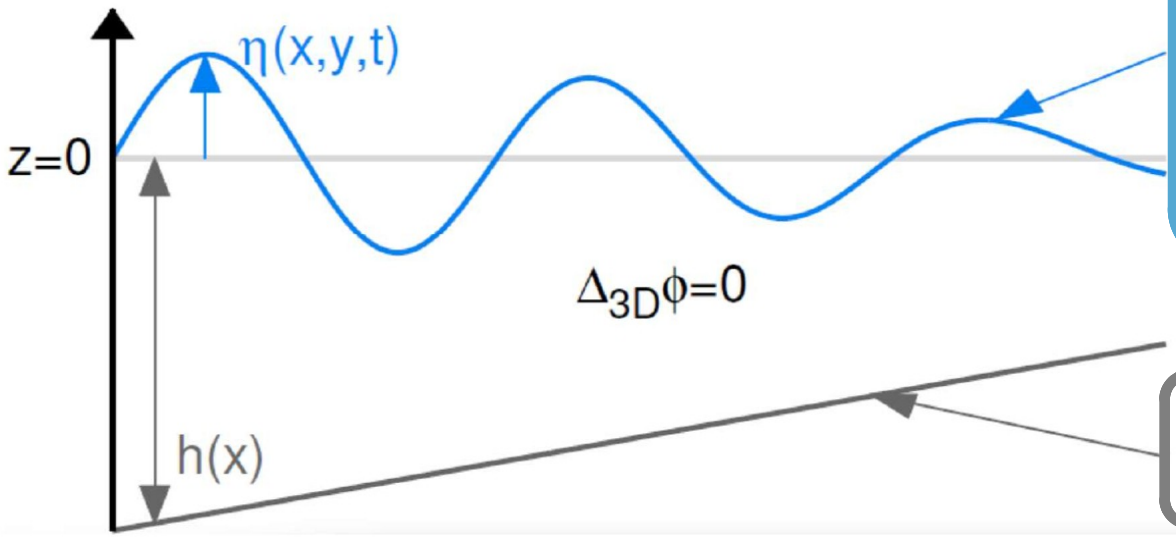
$$\frac{\partial \eta}{\partial t} + \nabla_{\mathrm{H}}\Phi.\nabla_{\mathrm{H}}\eta - \frac{\partial \Phi}{\partial z} = 0$$

$$\frac{\partial \Phi}{\partial t} + \frac{1}{2}(\nabla\Phi)^2 + g\eta = 0 \qquad \text{at } z = \eta(x, y, t)$$

$$\frac{\partial h}{\partial t} + \nabla_{\mathrm{H}}\Phi.\nabla_{\mathrm{H}}h + \frac{\partial \Phi}{\partial z} = 0 \qquad \text{at } z = -h(x, y, t)$$

Nonlinear potential flow problem

- System of 4 equations to solve :
  - Conservation of mass
  - Kinematic free surface boundary condition
  - Dynamic free surface boundary condition
  - Bottom boundary condition



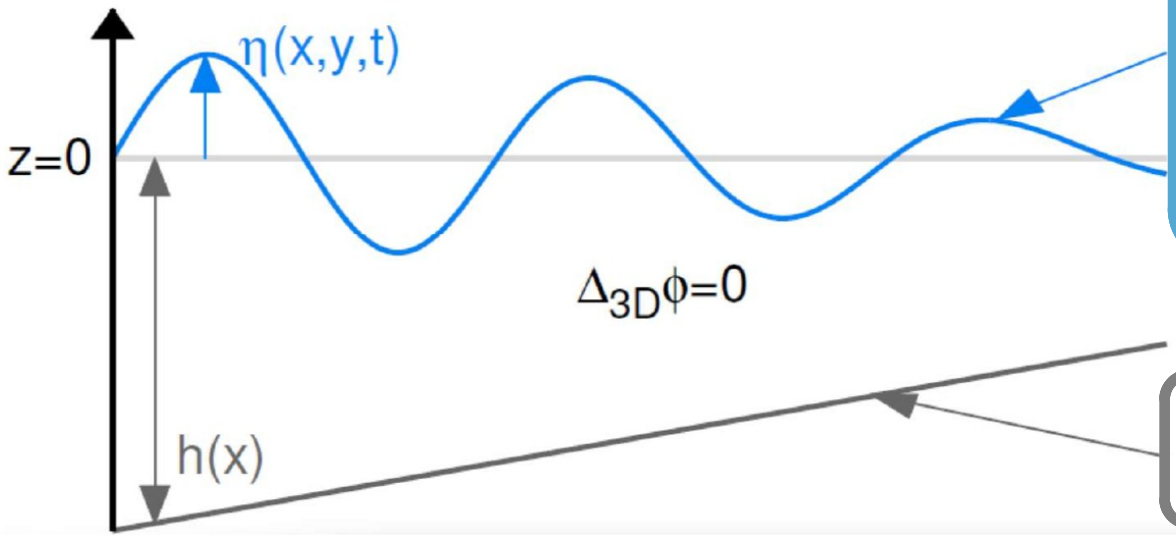
$$\frac{\partial \eta}{\partial t} + \nabla_H \Phi . \nabla_H \eta - \frac{\partial \Phi}{\partial z} = 0$$

$$\frac{\partial \Phi}{\partial t} + \frac{1}{2}(\nabla \Phi)^2 + g\eta = 0$$

$$\frac{\partial h}{\partial t} + \nabla_H \Phi . \nabla_H h + \frac{\partial \Phi}{\partial z} = 0$$

Nonlinear potential flow problem

- Challenge :
  - Nonlinear system of equations
  - Applied at free surface boundary, η, which is unknown



$$\frac{\partial \eta}{\partial t} + \nabla_H \Phi . \nabla_H \eta - \frac{\partial \Phi}{\partial z} = 0$$

$$\frac{\partial \Phi}{\partial t} + \frac{1}{2}(\nabla \Phi)^2 + g\eta = 0$$

$$\frac{\partial h}{\partial t} + \nabla_H \Phi . \nabla_H h + \frac{\partial \Phi}{\partial z} = 0$$

Linearize the system of equations

- Assume small amplitude waves
  - Relative to the wavelength: $H/L \ll 1$
  - Relative to the water depth: $H/h \ll 1$
- Evaluate scaling of each term
- Eliminate hidden nonlinearity in applying the boundary condition at $z=\eta$, by doing a Taylor series expansion at $z=0$

Nonlinear potential flow problem

$$\Delta\Phi = 0 \qquad \text{in } \Omega$$

$$\frac{\partial \eta}{\partial t} + \nabla_{\mathrm{H}}\Phi \cdot \nabla_{\mathrm{H}}\eta - \frac{\partial \Phi}{\partial z} = 0$$

$$\frac{\partial \Phi}{\partial t} + \frac{1}{2}(\nabla\Phi)^2 + g\eta = 0 \qquad \text{at } z = \eta(x, y, t)$$

$$\frac{\partial h}{\partial t} + \nabla_{\mathrm{H}}\Phi \cdot \nabla_{\mathrm{H}}h + \frac{\partial \Phi}{\partial z} = 0 \qquad \text{at } z = -h(x, y, t)$$

Linear potential flow problem

$$\Delta\Phi = 0 \qquad \text{in } \Omega$$

$$\frac{\partial \eta}{\partial t} = \tilde{w}_0 \qquad \text{at } z = 0$$

$$\frac{\partial \tilde{\Phi}_0}{\partial t} = -g\eta \qquad \text{at } z = 0$$

$$\frac{\partial \Phi}{\partial z} = 0 \qquad \text{at } z = -h$$

Linearized (Airy) wave theory

$$\frac{\partial^2 \phi}{\partial t^2} + g\frac{\partial \phi}{\partial z} = 0, \quad z = 0$$



$$\nabla^2 \phi = \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial z^2} = 0$$

Search for solutions that are periodic waves of permanent form



$$\frac{\partial \phi}{\partial n} = \frac{\partial \phi}{\partial z} = 0, \quad z = -h$$

Linearized (Airy) wave theory

- Waves of permanent form travelling with constant speed c, defined as c=ω/k:
  - ω = 2π/T, wave frequency
  - k = 2π/L, wave number
- Using separation of variables, with $\theta = x - ct$,

$$\phi = Z(z)F(\theta)$$

- And substituting in the Laplace equation:

$$\frac{Z''(z)}{Z(z)} = -\frac{F''(\theta)}{F(\theta)}$$

Linearized (Airy) wave theory

- Separation constant (units L^{-2}) chosen to be k^2:

$$Z'' - k^2 Z = 0$$

$$Z(z) = A_1 \exp(kz) + A_2 \exp(-kz)$$

$$Z(z) = A_3 \cosh(k(z+h))$$

$$F'' + k^2 F = 0$$

$$F(\theta) = B_1 \cos(kx - \omega t) + B_2 \sin(kx - \omega t)$$

$$F(\theta) = B_3 \sin(kx - \omega t + \delta)$$

Apply at bottom boundary

$$\phi = A \cosh(k(z+h)) \sin(kx - \omega t)$$

arbitrary constant

Linearized (Airy) wave theory

- Separation constant (units L^{-2}) chosen to be k^2:

$$Z'' - k^2 Z = 0$$

$$Z(z) = A_1 \exp(kz) + A_2 \exp(-kz)$$

$$Z(z) = A_3 \cosh(k(z+h))$$

$$F'' + k^2 F = 0$$

$$F(\theta) = B_1 \cos(kx - \omega t) + B_2 \sin(kx - \omega t)$$

$$F(\theta) = B_3 \sin(kx - \omega t + \delta)$$

Apply at bottom boundary

$$\phi = A \cosh(k(z+h)) \sin(kx - \omega t)$$

arbitrary constant

$$\frac{\partial \tilde{\Phi}_0}{\partial t} = -g\eta$$

Calculate the free surface profile using the dynamic free surface boundary condition

Linearized (Airy) wave theory

- Corresponding free surface profile:

$$\eta = \frac{A\omega}{g}\cosh(kh)\cos(kx - \omega t)$$

- Defining the wave amplitude a=H/2 and solving for A:

$$\eta = a\cos(kx - \omega t)$$

$$A = \frac{ag}{\omega}\frac{1}{\cosh(kh)}$$

- Velocity potential for a linear progressive wave:

$$\phi = \frac{ag}{\omega}\frac{\cosh(k(z+h))}{\cosh(kh)}\sin(kx - \omega t)$$

Linear dispersion relation

- Substitute the linear potenial wave solution:

$$\phi = \frac{ag}{\omega}\frac{\cosh(k(z+h))}{\cosh(kh)}\sin(kx-\omega t)$$

- in the combined free surface boundary condition:

$$\frac{\partial^2\phi}{\partial t^2} + g\frac{\partial\phi}{\partial z} = 0, \quad z = 0$$

Linear dispersion relation

- Apply at the free surface and rearrange to solve for the frequency:

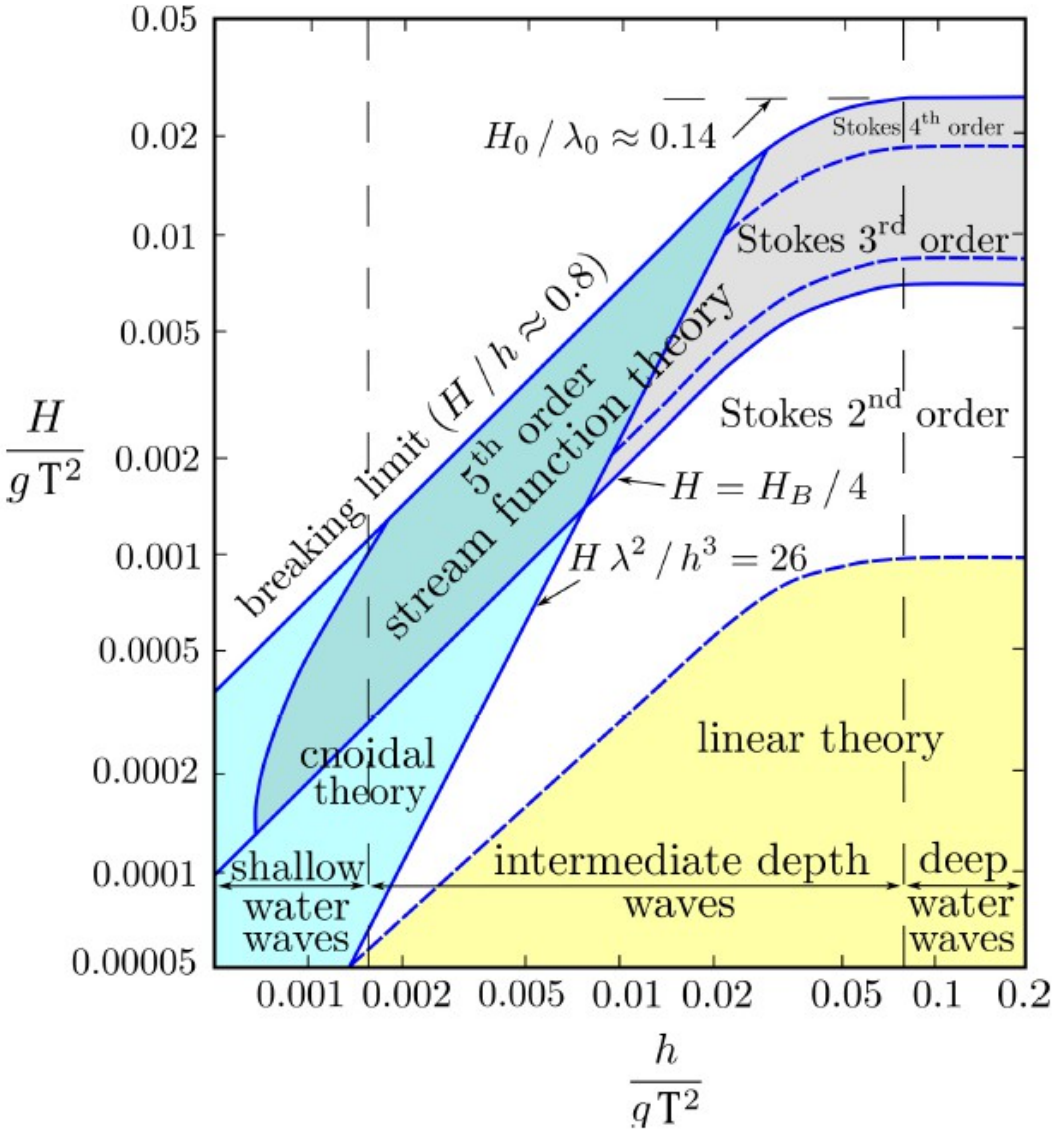
$$\frac{\partial^2 \phi}{\partial t^2} + g\frac{\partial \phi}{\partial z} = 0, \quad z = 0$$

$$-\omega^2 \phi_{z=0} + gk\frac{\sinh(kh)}{\cosh(kh)}\phi_{z=0} = 0$$

$$\omega^2 = gk\tanh(kh)$$

Important: defines the relationship between the wave number (or wavelength) and the wave frequency (or period)

Nonlinear dispersion relation



? ?

$\omega^2 = gk \tanh(kh)$

Nonlinear dispersion relation

- Stokes 5th order dispersion relation:

$$\frac{\omega}{\omega_0} = 1 + k^2 a^2 \omega_2 + k^4 a^4 \omega_4,$$

$$\omega_0 = \sqrt{gk\sigma}, \quad \omega_2 = \frac{2\alpha_1^2 + 7}{4(\alpha_1 - 1)^2} \quad \omega_4 = \frac{20\alpha_1^5 + 112\alpha_1^4 - 100\alpha_1^3 - 68\alpha_1^2 - 211\alpha_1 + 328}{32(\alpha_1 - 1)^5}.$$

$$\sigma = \tanh kh, \quad \alpha_1 = \cosh 2kh = \frac{1 + \sigma^2}{1 - \sigma^2},$$

(Zhao et Liu, 2021)

Important: defines the relationship between the wave number (or wavelength) and the wave frequency (or period)

Linear dispersion relation

$$\omega^2 = gk \tanh(kh)$$

- Explicit relationship for ω=f(k,h)
- Implicit relationship for k=f(ω,h)

- Most often have measurements of ω or T (wave buoy)
- Methods for solving the implicit dispersion relation:
  - Graphical representation
  - Explicit approximations
  - Numerical iteration

Solving the dispersion relation

1. Numerical iteration:

$$k_n = \frac{\omega^2}{g\,tanh(\,k_{n-1}h\,)}$$

```matlab
% calculate dispersion relation
dk=1
while dk>0.00005
    k = omega^2/(g*tanh(k0*h));
    dk = abs(k-k0);
    k0 = k;
    L = 2*pi/k
end
```

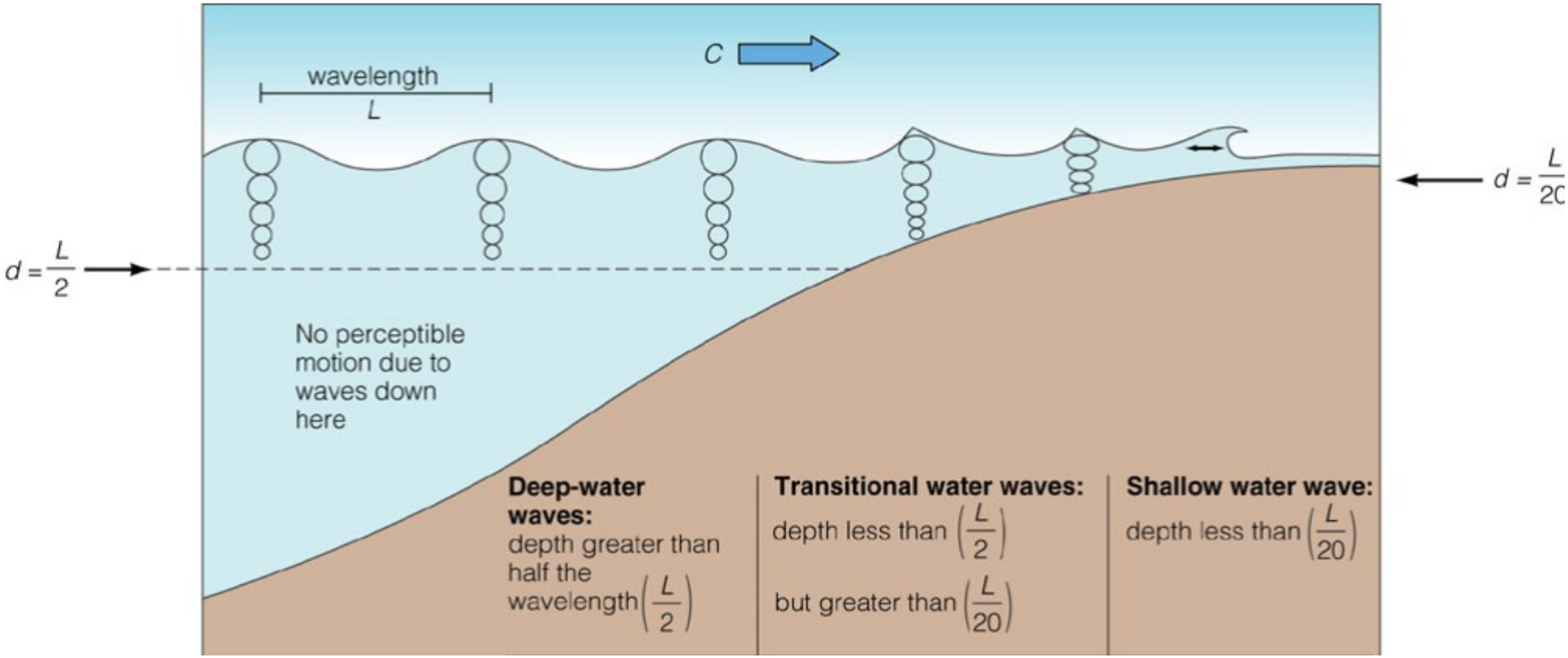
Matlab

```python
def k_dispersion(omega):
    omega_thresh=10**(-5)
    k=1.
    omegadiff=1.
    counter=0
    while np.abs(omegadiff)>omega_thresh :
        counter=counter+1
        omega_est=np.sqrt((g*k*np.tanh(k*h)))
        omegadiff=omega_est-omega
        domegadk=(g*k*h/(np.cosh(k*h))**2+g*np.tanh(k*h))/(2.*omega)
        kdiff=omegadiff/domegadk
        k=k-kdiff
    return k
```

Python

Shallow and deep water limits

- Relative water depth: kh or h/L
- Nondimensional wave period: $T\sqrt{g/h}$



http://science.kennesaw.edu/~jdirnber/oceanography/LecturesOceanogr/LecWaves/

Approximations in shallow or deep water

$$\omega^2 = gk \tanh(kh)$$

Variable	Approximations with criteria	
	Shallow-water or long wave approximation (*small kh*)	Deep-water or short wave approximation (*large kh*)
	h/*L* < 1/25 or $T\sqrt{g/h} > 25$	*h*/*L* > 1/2 or $T\sqrt{g/h} < 4$
Dispersion relation	$\omega^2 = gh\,k^2$	$\omega^2 = gk_o$
Wave number *k* (rad/m)	$k = \omega/\sqrt{gh}$	$k_o = \omega^2/g$
Wavelength *L* (m)	$L = T\sqrt{gh}$	$L_o = gT^2/(2\pi)$
Phase speed *c* (m/s)	$c = \sqrt{gh}$	$c_o = gT/(2\pi)$

* All based on linear wave theory

Approximations in shallow or deep water

$$\omega^2 = gk \tanh(kh)$$

Variable	Approximations with criteria
	Shallow-water or long wave approximation (*small kh*)
	h/*L* < 1/25 or $T\sqrt{g/h} > 25$
Dispersion relation	$\omega^2 = gh\,k^2$
Wave number *k* (rad/m)	$k = \omega/\sqrt{gh}$
Wavelength *L* (m)	$L = T\sqrt{gh}$
Phase speed *c* (m/s)	$c = \sqrt{gh}$

Wave phase speed depends only on the water depth



* All based on linear wave theory

Approximations in shallow or deep water

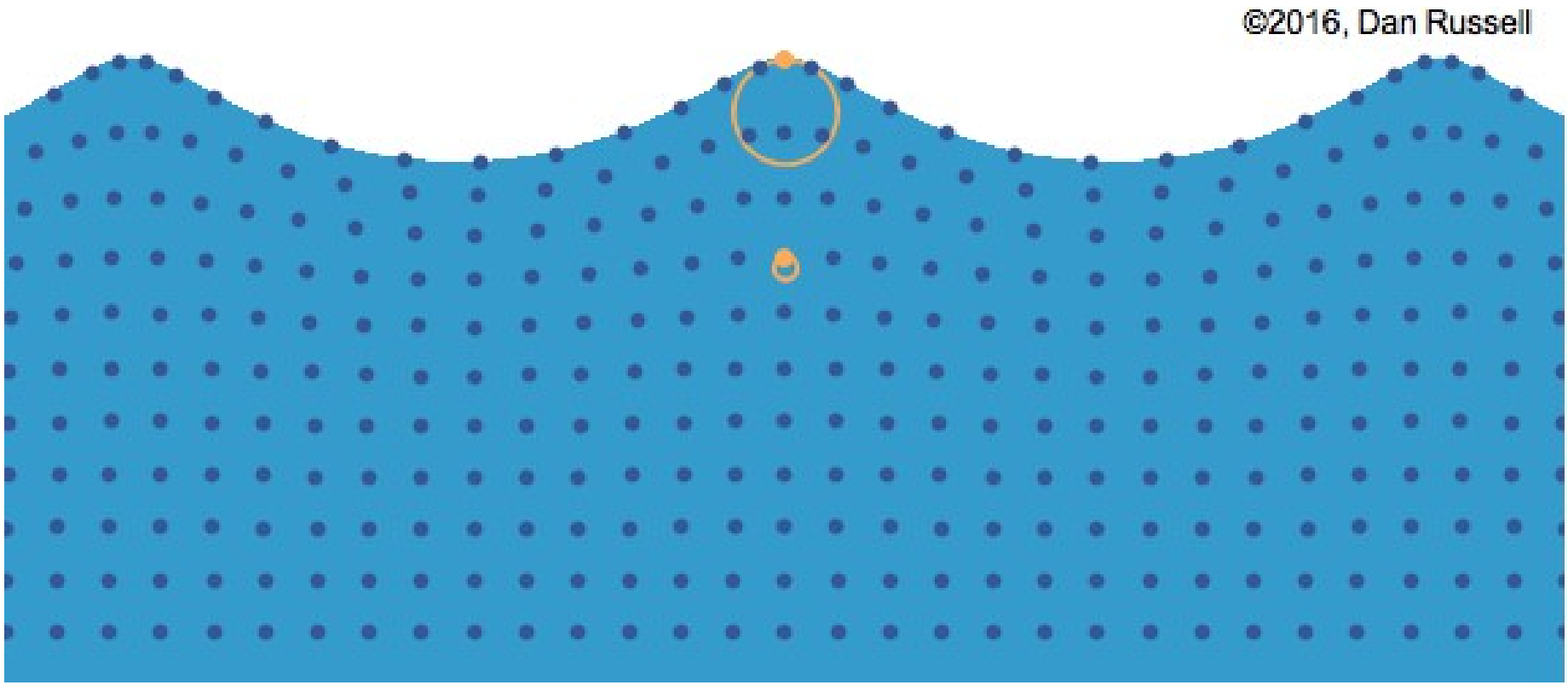
$$\omega^2 = gk \tanh(kh)$$

Variable	Approximations with criteria
	Deep-water or short wave approximation (*large kh*)
	h/*L* > 1/2 or $T\sqrt{g/h} < 4$
Dispersion relation	$\omega^2 = gk_o$
Wave number *k* (rad/m)	$k_o = \omega^2/g$
Wavelength *L* (m)	$L_o = gT^2/(2\pi)$
Phase speed *c* (m/s)	$c_o = gT/(2\pi)$

Independent of the water depth

* All based on linear wave theory

Wave orbital velocities

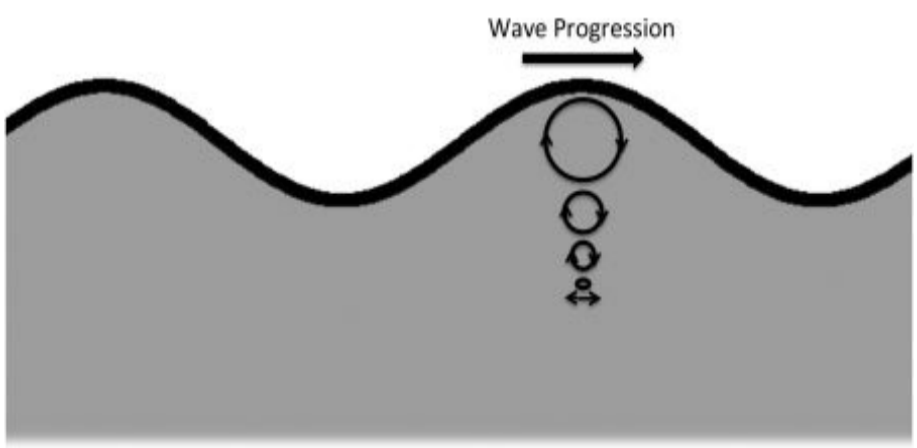


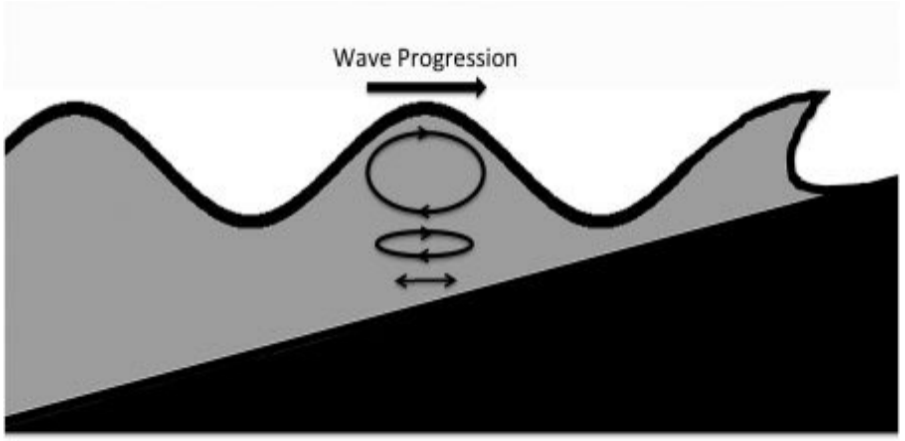
Wave orbital velocities

- Horizontal and vertical velocity:

$$u(x,z,t) = \frac{\partial \phi}{\partial x} = \frac{\pi H}{T} \frac{\cosh(k(z+h))}{\sinh(kh)} \cos(kx - \omega t)$$

$$w(x,z,t) = \frac{\partial \phi}{\partial z} = \frac{\pi H}{T} \frac{\sinh(k(z+h))}{\sinh(kh)} \sin(kx - \omega t)$$





Fish and Hoffman (2015)

Wave orbital velocities

- Horizontal and vertical velocity:

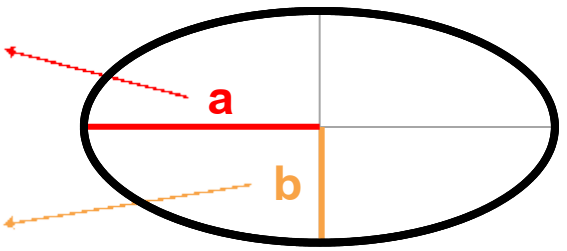
$$u(x,z,t) = \frac{\partial \phi}{\partial x} = \frac{\pi H}{T} \frac{\cosh(k(z+h))}{\sinh(kh)} \cos(kx - \omega t)$$

$$w(x,z,t) = \frac{\partial \phi}{\partial z} = \frac{\pi H}{T} \frac{\sinh(k(z+h))}{\sinh(kh)} \sin(kx - \omega t)$$

- Elliptical particle trajectories:

Major axis: $a = \frac{gHk}{2\omega^2} \frac{\cosh(k(z_0+h))}{\cosh(kh)}$

Minor axis: $b = \frac{gHk}{2\omega^2} \frac{\sinh(k(z_0+h))}{\cosh(kh)}$



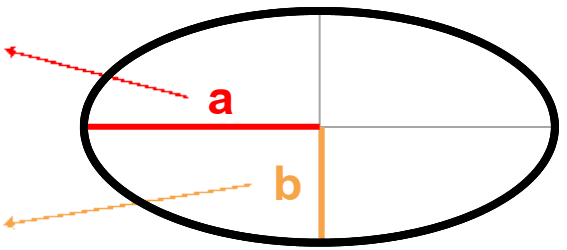
Wave orbital velocities

- Quick calculation of maximum horizontal particle trajectory conditions:
  - H = 2 m
  - T = 10 s
  - h = 1500 m

- Elliptical particle trajectories:

Major axis: $$a = \frac{gHk}{2\omega^2}\frac{\cosh(k(z_0+h))}{\cosh(kh)}$$

Minor axis: $$b = \frac{gHk}{2\omega^2}\frac{\sinh(k(z_0+h))}{\cosh(kh)}$$

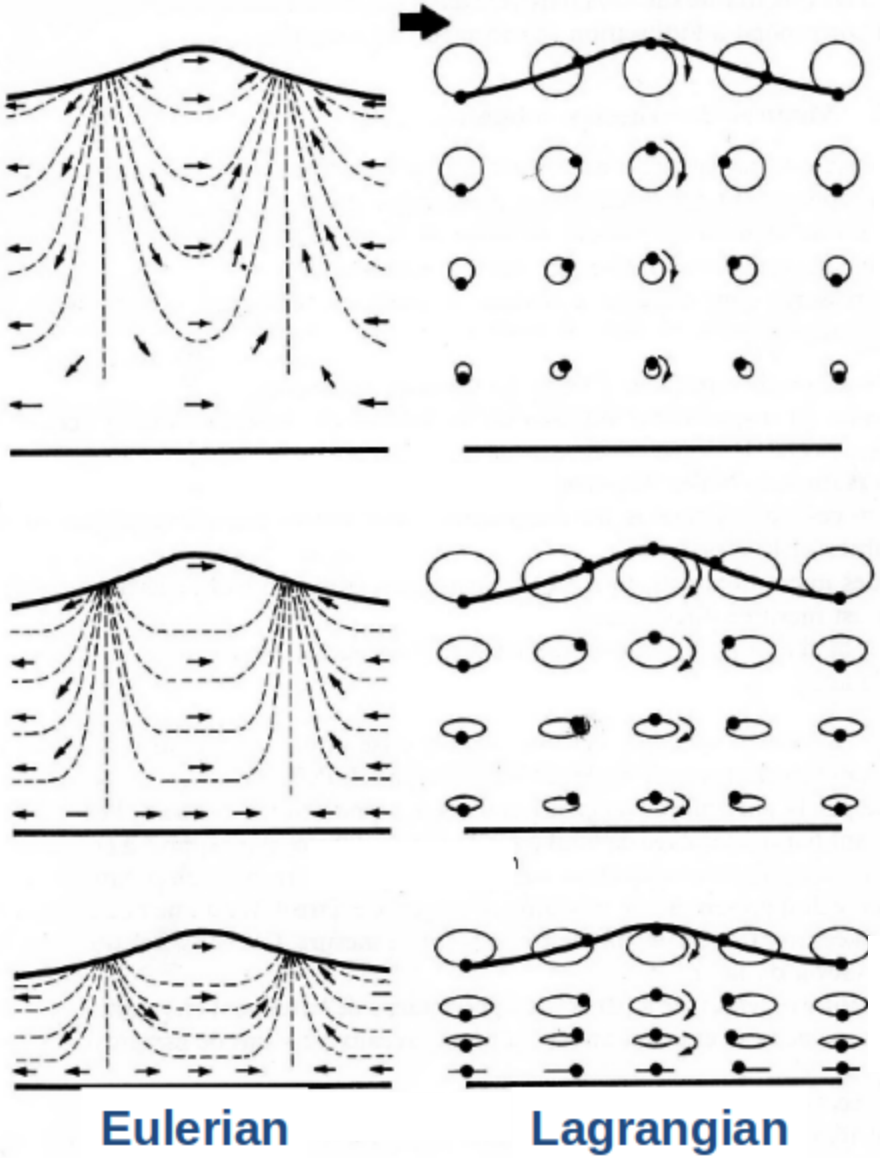


Wave orbital velocities

Deep water
(waves don't « feel » the bottom)

Transition zone

Shallow water



Circular orbits

Elliptical orbits

Pressure

- Bernoulli equation:

$$\frac{p}{\rho} + gz + \frac{\partial \phi}{\partial t} + \frac{(\nabla \phi)^2}{2} = 0$$

(linearize)

- For linear waves:

$$p = -\rho g z - \rho \frac{\partial \phi}{\partial t}$$

$$= -\rho g z + \rho \frac{gH}{2} \frac{\cosh(k(z+h))}{\cosh(kh)} \cos(kx - \omega t)$$

which can be rewritten as:

$$p = -\rho g z - \rho g \eta K_p(z) \quad \text{with} \quad K_p(z) = \frac{\cosh(k(z+h))}{\cosh(kh)} \leq 1$$

Pressure

- Bernoulli equation:

$$\frac{p}{\rho} + gz + \frac{\partial \phi}{\partial t} + \frac{(\nabla \phi)^2}{2} = 0$$

- For linear waves:

$$p = -\rho g z - \rho \frac{\partial \phi}{\partial t}$$

$$= -\rho g z + \rho \frac{gH}{2} \frac{\cosh(k(z+h))}{\cosh(kh)} \cos(kx - \omega t)$$

Hydrostatic pressure

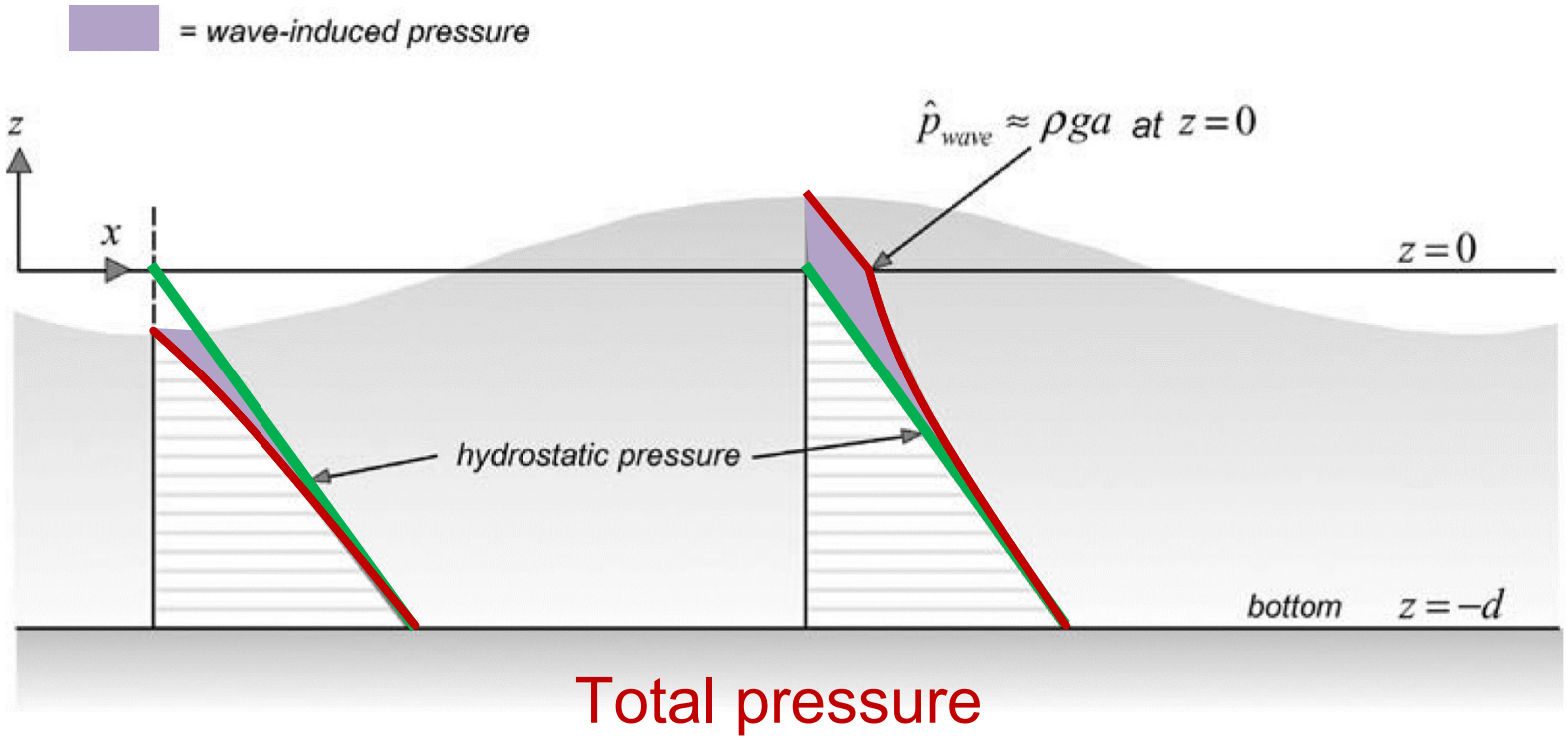
$$p = -\rho g z - \rho g \eta K_p(z)$$

with $K_p(z) = \frac{\cosh(k(z+h))}{\cosh(kh)} \leq 1$

Dynamic pressure

Pressure

- Hydrostatic and dynamic (or wave-induced) pressure under wave crests and troughs

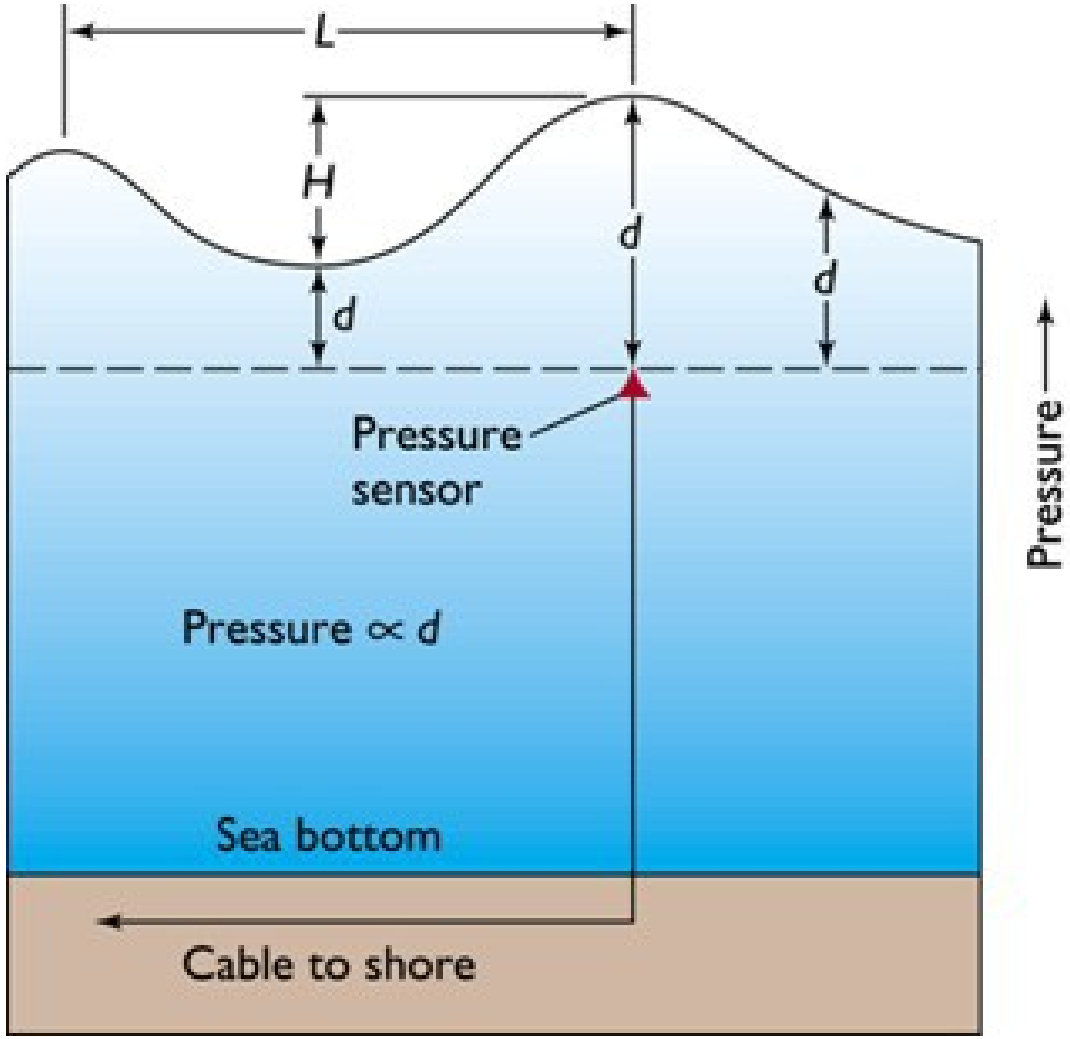


(Holthuijsen, 2007)

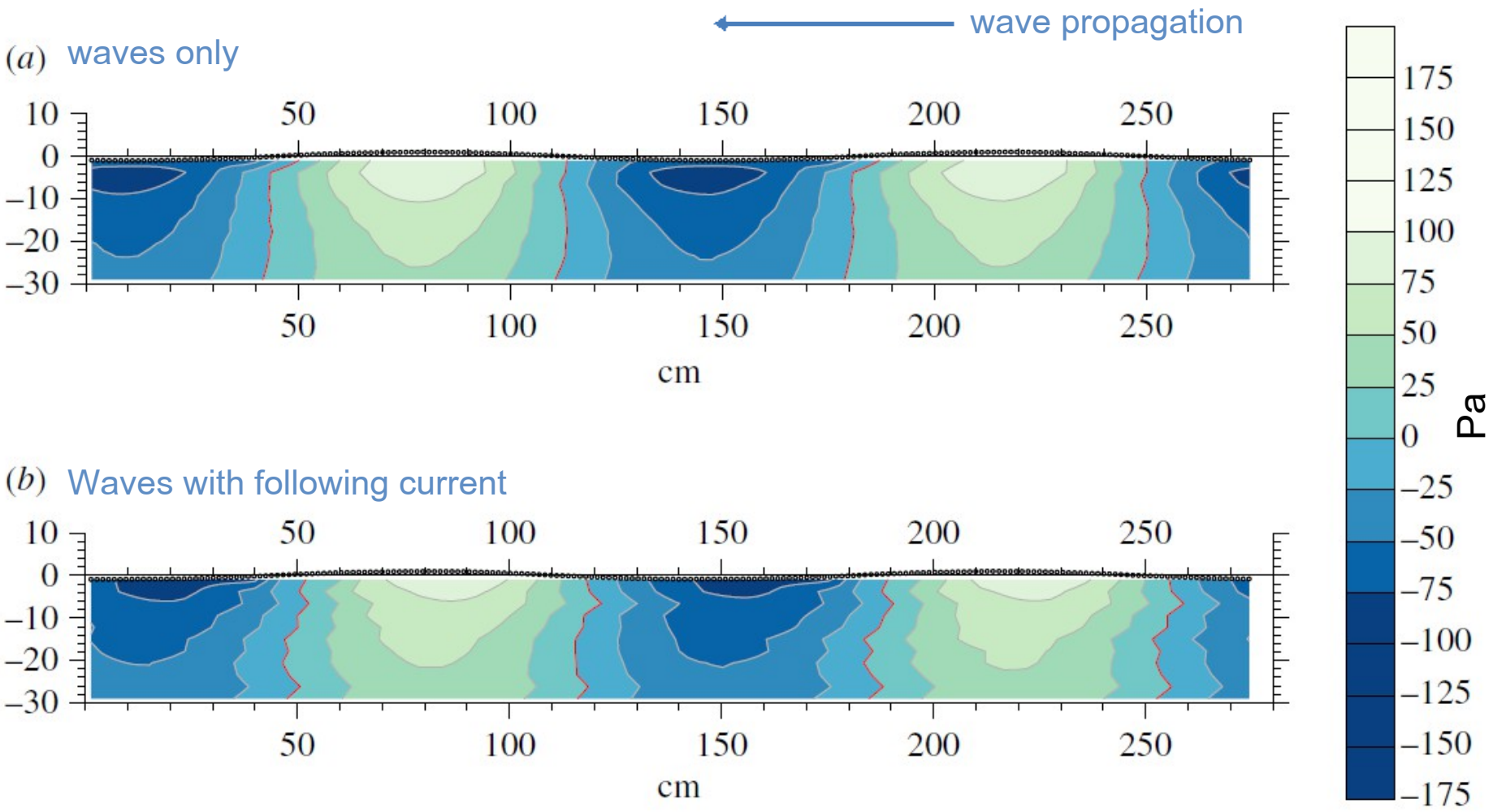
Pressure

- **Implications:** allows measuring wave height using pressure sensors

- **Assumptions:** pressure changes with passing wave (hydrostatic, or nonhydrostatic with nonlinear correction term)



Pressure



Umeyama (2017)

Pressure reconstruction

$$p = -\rho g z - \rho g \eta K_p(z)$$

Total pressure = Hydrostatic pressure + Dynamic pressure

with $K_p(z) = \dfrac{\cosh(k(z+h))}{\cosh(kh)} \leq 1$

- Used in laboratory and field experiments
- Linear theory:
  - ~5% errors in deep and intermediate wave (Bishop and Donelan, 1987)
  - ~30% errors in shallow water (Martins et al., 2017) due to nonlinear effects

Wave energy

- Potential energy: vertical deplacement of water

$$E_p = \int_{x=0}^{L} \int_{z=-h}^{\eta} \rho g z \, dz \, dx = \int_{x=0}^{L} \rho g \frac{\eta^2}{2} dx = \frac{\rho g H^2 L}{16}$$

- Kinetic energy: fluid movement

$$E_k = \int_{x=0}^{L} \int_{z=-h}^{\eta} \frac{1}{2} u^2 \, dz \, dx = \frac{\rho g H^2 L}{16}$$

Potential energy = kinetic energy

- Total energy per wave: potential + kinetic

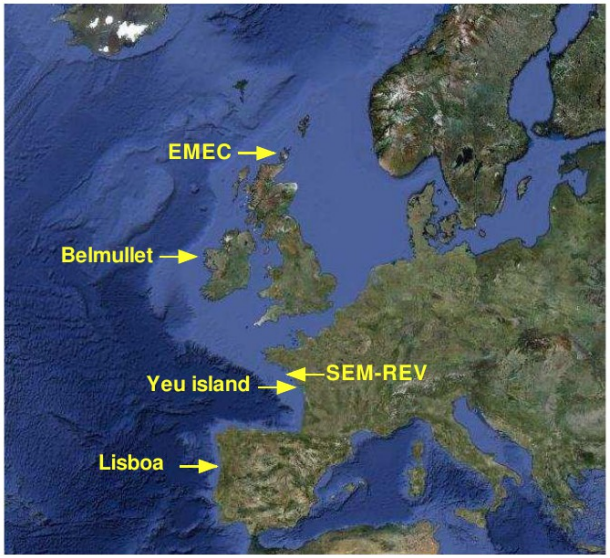
$$E = \frac{1}{L}\left(\frac{\rho g H^2 L}{16} + \frac{\rho g H^2 L}{16}\right) = \boxed{\frac{\rho g H^2}{8}}$$

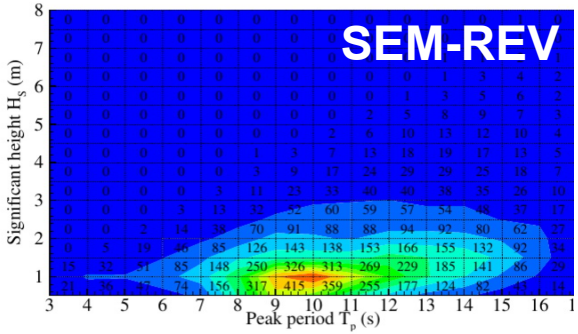
Wave energy converters

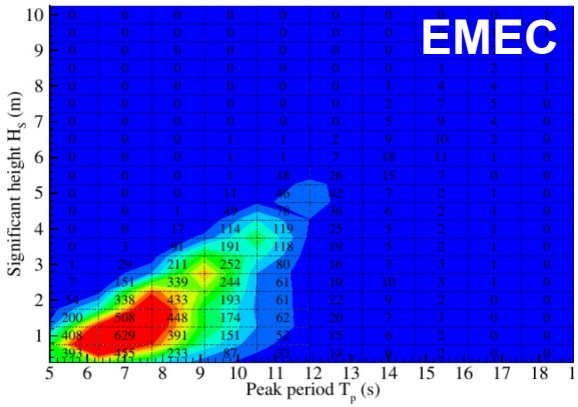
Numerical benchmarking study of a selection of Wave Energy Converters

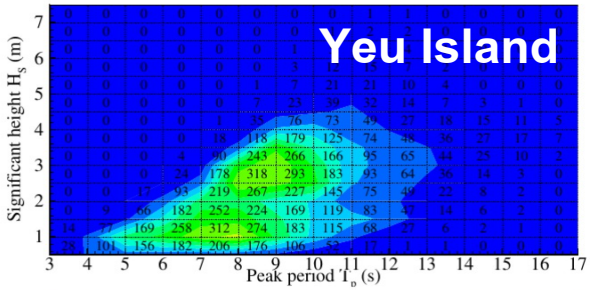
A. Babarit[a,*], J. Hals[b], M.J. Muliawan[b], A. Kurniawan[b], T. Moan[b], J. Krokstad[c]

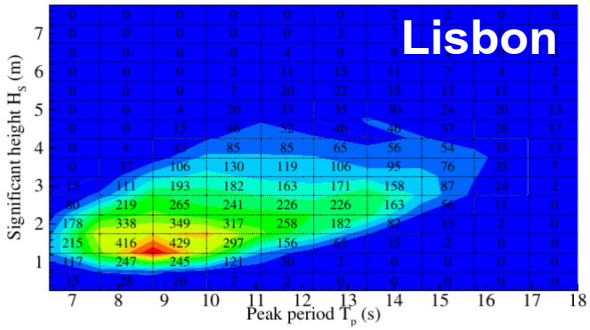
- Site characteristics

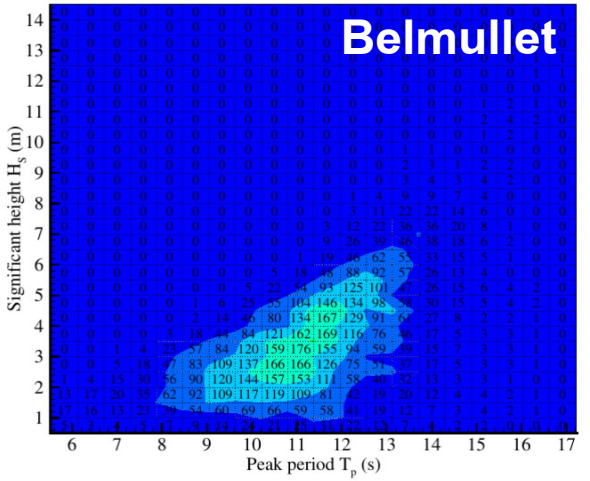












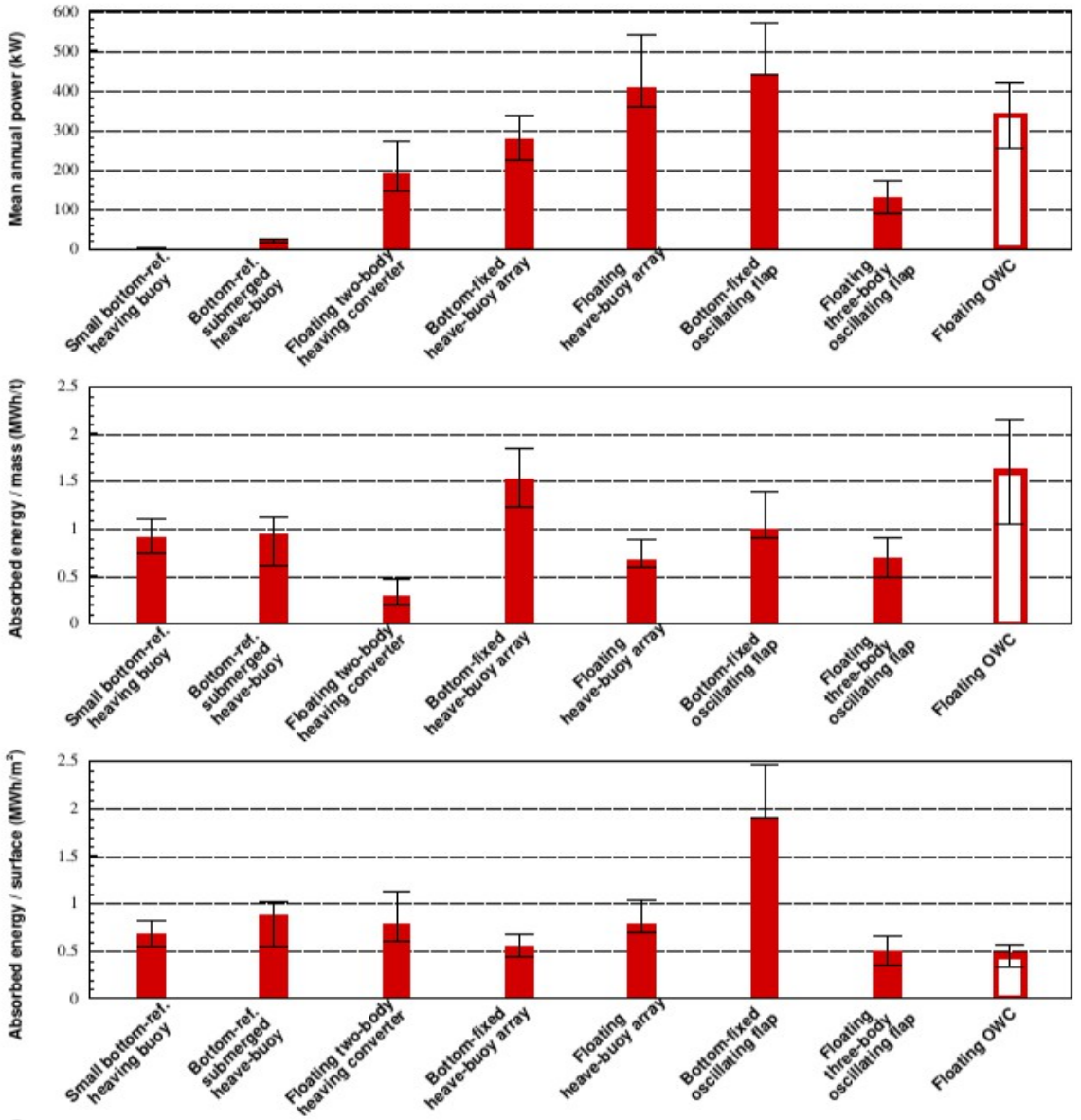
(Babarit et al., 2012)

Wave energy converters

Mean annual power

Absorbed energy/mass

Absorbed energy/surface area



(Babarit et al., 2012)

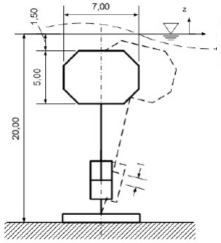
Wave energy converters

- Power matrices for 8 different WECs

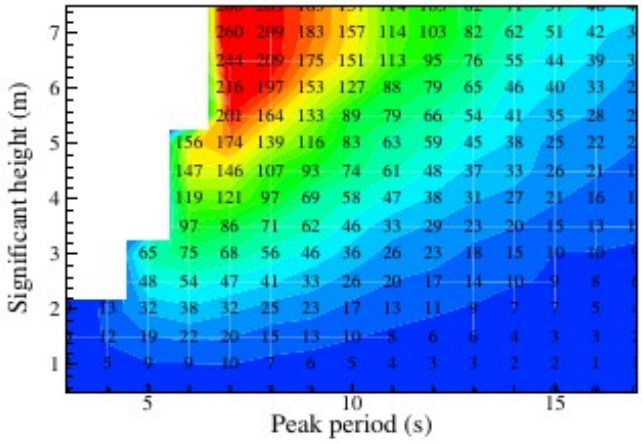


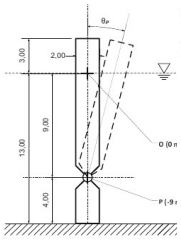




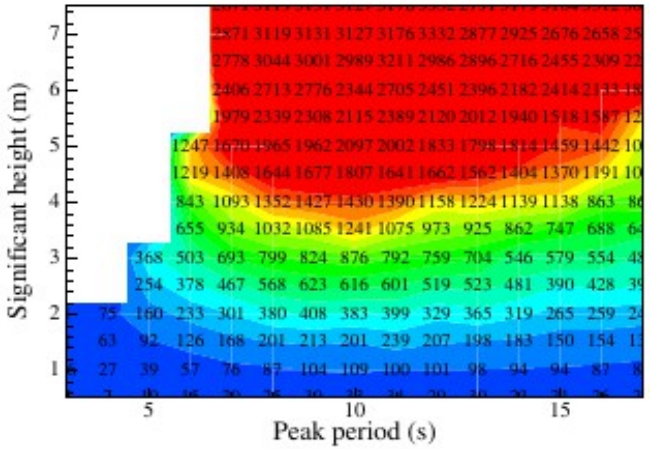


Power matrix





Power matrix



(Babarit et al., 2012)

Wave energy flux

- Average wave energy flux:

$$F_E = \frac{1}{T}\int_{t=0}^{T}\int_{z=-h}^{\eta} pu\,dz\,dt$$

$$F_E = \frac{1}{T}\int_{t=0}^{T}\int_{z=-h}^{0} \left(p_{dynamic} + \rho g z\right) u\,dz\,dt$$

is calculated as:

$$F_E = \underbrace{\frac{\rho g H^2}{8}}_{E}\ \underbrace{\left[\frac{1}{2}\left(1 + \frac{2kh}{\sinh(2kh)}\right)\right]}_{n}\ \underbrace{\frac{L}{T}}_{C}$$

Group velocity

- Wave group velocity: average speed of wave energy propagation:

$$C_g = \frac{F_E}{E} = \underbrace{\left[\frac{1}{2}\left(1 + \frac{2kh}{\sinh(2kh)}\right)\right]}_{n} \underbrace{C}_{C} = nC$$

where $1/2 \leq n \leq 1$, such that $C_g \leq C$.

Wave energy propagates slower than individual waves.



https://en.wikipedia.org/wiki/Dispersion_%28water_waves%29#/media/File:Wave_group.gif

Group velocity

- Wave group velocity: average speed of wave energy propagation:

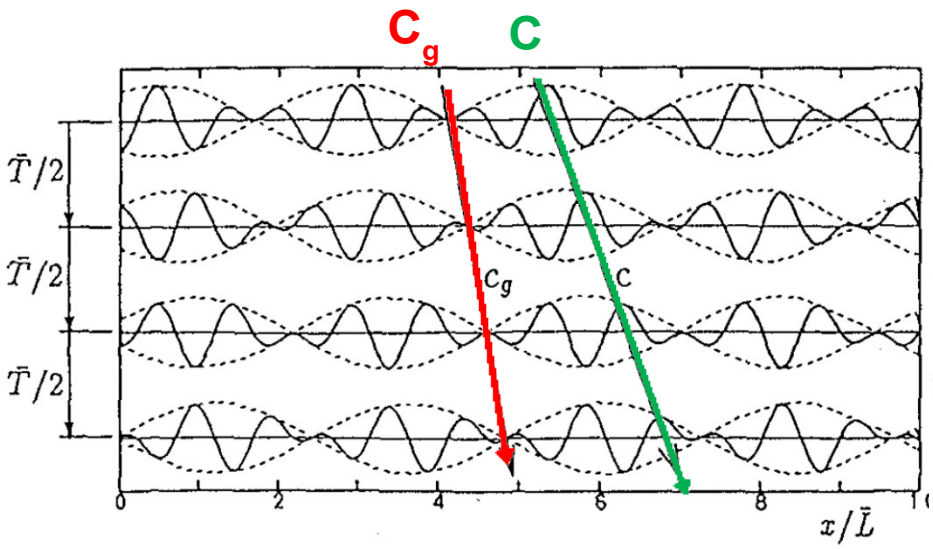
$$C_g = \frac{F_E}{E} = \underbrace{\left[\frac{1}{2}\left(1 + \frac{2kh}{\sinh(2kh)}\right)\right]}_{n} \underbrace{C}_{C} = nC$$

where $1/2 \leq n \leq 1$, such that $C_g \leq C$.

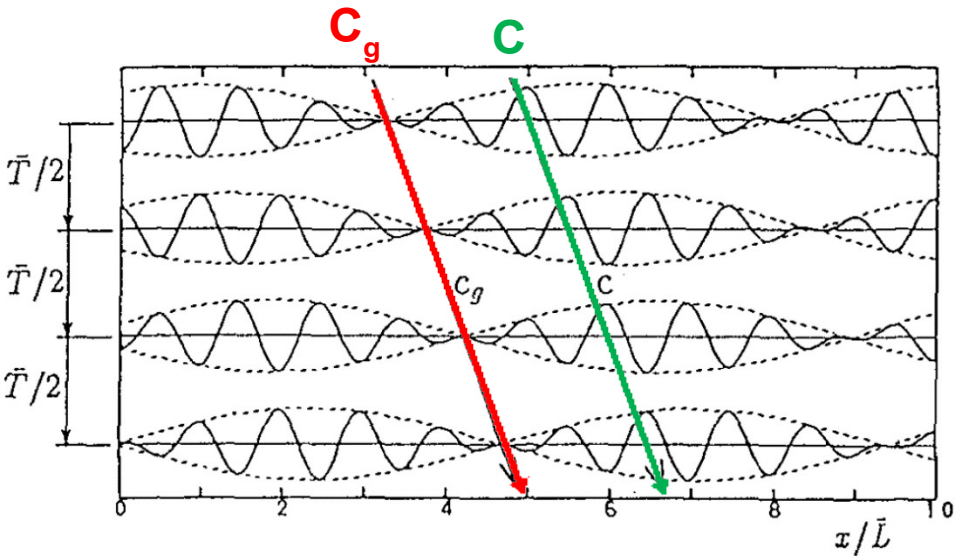
- Shallow water limit: $C_g = C = \sqrt{gh}$
- Deep water limit: $C_g = \frac{C}{2}$

Group velocity

Example: superposition of two sinusoidal waves with the same wave height H and frequencies $0.9\bar{f}$ and $1.1\bar{f}$



Deep water = dispersive case



Shallow water = non-dispersive case

(Kubo et al. ICCE 1992)

Shallow water and deep water approximations for linear wave theory

Variable	Approximations with criteria	
	Shallow-water or long wave approximation (*small kh*)	Deep-water or short wave approximation (*large kh*)
	h/*L* < 1/25 or $T\sqrt{g/h} > 25$	*h*/*L* > 1/2 or $T\sqrt{g/h} < 4$
Dispersion relation	$\omega^2 = gh\,k^2$	$\omega^2 = gk_o$
Wave number *k* (rad/m)	$k = \omega/\sqrt{gh}$	$k_o = \omega^2/g$
Wavelength *L* (m)	$L = T\sqrt{gh}$	$L_o = gT^2/(2\pi)$
Phase speed *c* (m/s)	$c = \sqrt{gh}$	$c_o = gT/(2\pi)$
Group velocity c_g (m/s)	$c_g = c = \sqrt{gh}$	$c_{go} = ½\,c_o = gT/(4\pi)$

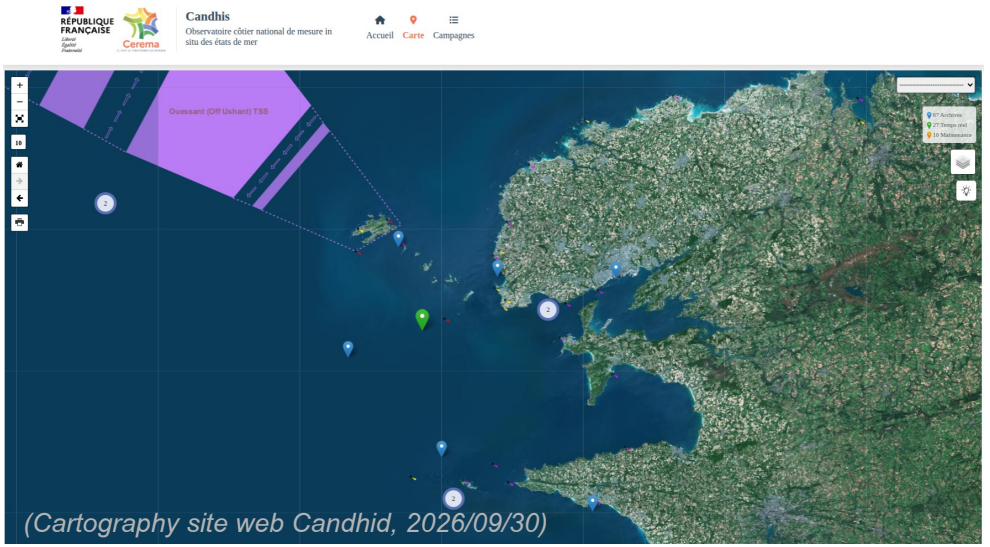
Exercise:

calculate wave characteristics

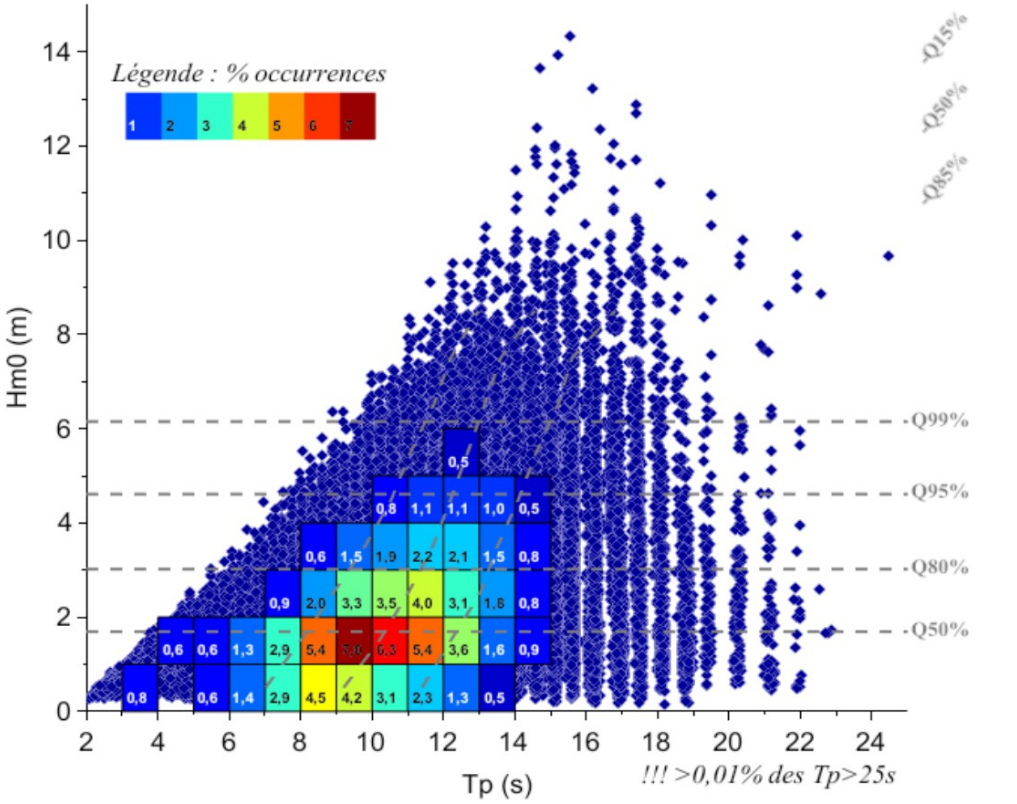
- Buoy : Les Pierres Noires
- What is the …
  - length of time used to calculate the sea state?
  - water depth ?
  - most frequently occurring wave height and period ?
  - today's wave height and period
  - wavelength ?
  - phase and group velocity ?

(deep water? shallow water?)

Candhis database
(https://candhis.cerema.fr/)

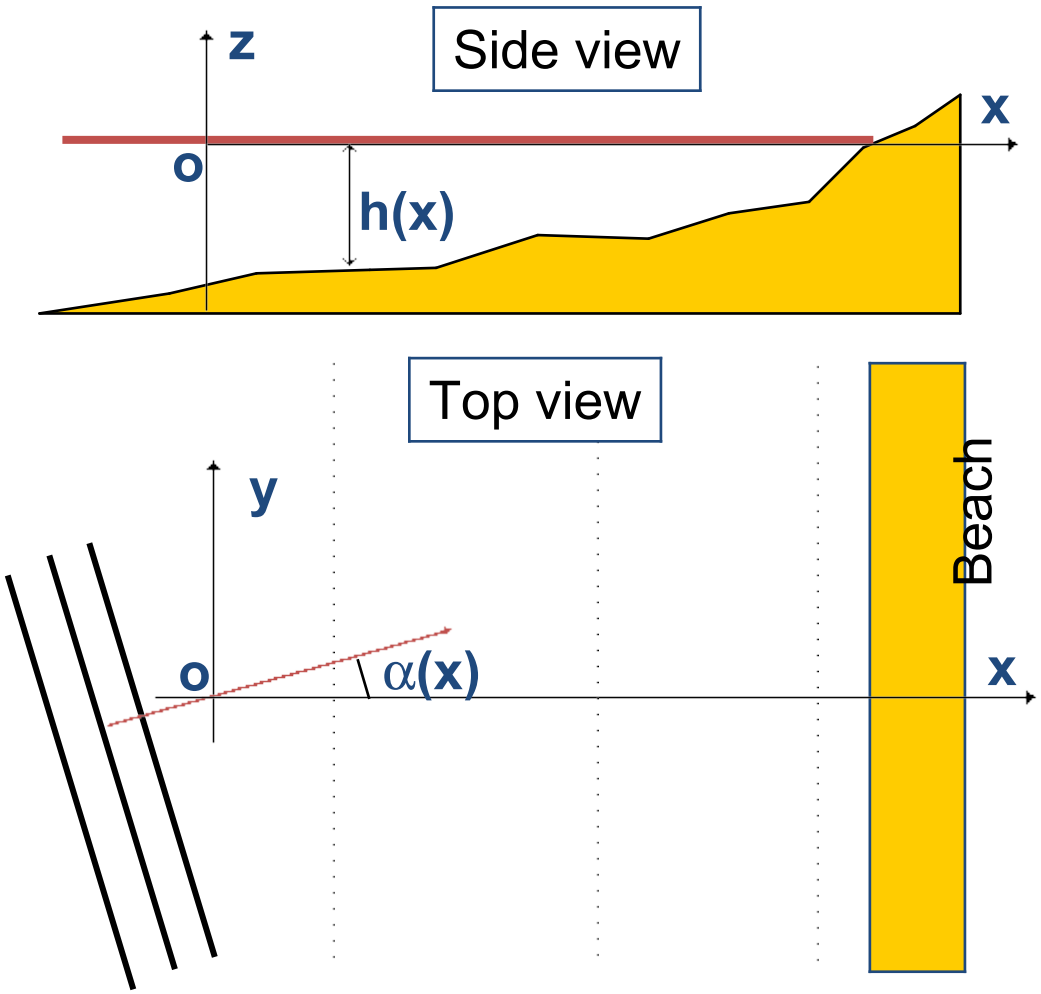


(Cartography site web Candhid, 2026/09/30)



(Extrait de Candhis : analyses 2026 des états de mer, Tome 1)

Bathymetric variations



Assumptions:
- linear waves
- monochromatic (T) and unidirectional (α)
 conservation of energy
- apply Airy theory locally
- alongshore homogeneous

Cross-shore variations of :
T, L (or k), C, C_g, H, α ?

Bathymetric variations

Example:



What do you observe as the water depth decreases?

- T, L, C, H?

Bathymetric variations

- Wave period does not change: **T or ω=2π/T constant**
- How do wavelength (L or wave number k) and wave celerity C vary?
  - Dispersion relation: $\omega^2 = gk\tanh(kh)$, with ω constant, k depends on h:

$$C' \equiv \left.\frac{\partial C}{\partial h}\right|_T = \frac{\partial}{\partial h}\left[\frac{g}{\omega}\tanh(kh)\right] = \frac{\partial}{\partial h}\left[\frac{gT}{2\pi}\tanh\left(\frac{2\pi h}{CT}\right)\right]$$

$$C' = \frac{gT}{2\pi}\frac{1}{\cosh^2(kh)}\frac{2\pi}{T}\frac{C - hC'}{C^2}$$

$$C' \equiv \left.\frac{\partial C}{\partial h}\right|_T = \frac{2\omega}{2kh + \sinh(2kh)} > 0$$

Bathymetric variations

Such that:

$$\left.\frac{\partial L}{\partial h}\right|_T = \left.\frac{\partial CT}{\partial h}\right|_T = T\left.\frac{\partial C}{\partial h}\right|_T = \frac{4\pi}{2kh + \sinh(2kh)} > 0$$

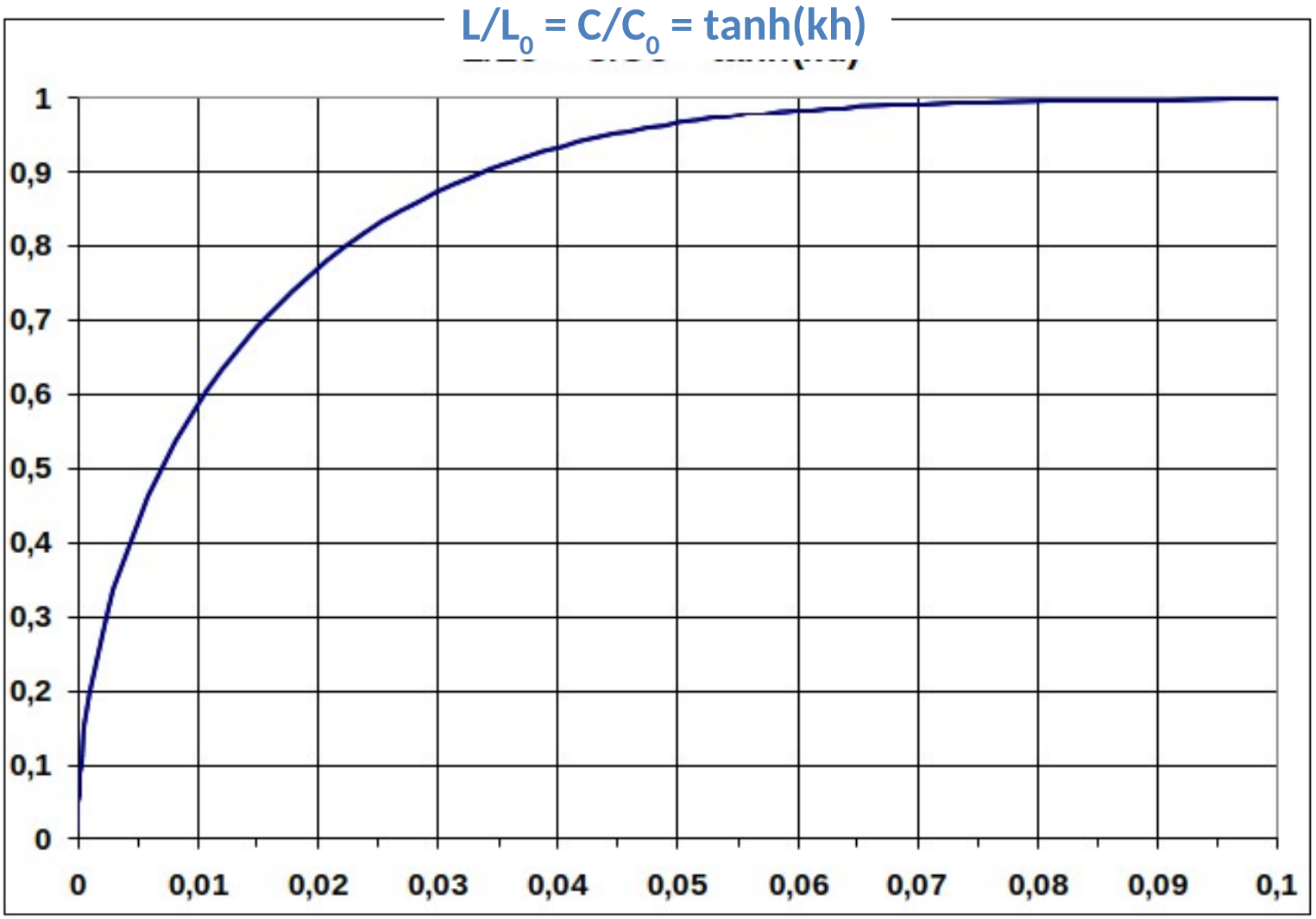
$$\left.\frac{\partial k}{\partial h}\right|_T = \left.\frac{\partial(\omega / C)}{\partial h}\right|_T = -\frac{\omega}{C^2}\left.\frac{\partial C}{\partial h}\right|_T < 0$$

As the water depth **h decreases**:

- Wave period does not change: **T or ω=2π/T constant**
- Wave length L **decreases**
- Wave celerity **C decreases**

(L and C decrease monotonically with a monotonic decrease in h)

Variation of $L/L_o = C/C_o = k_o/k$



$$\frac{h}{gT^2}$$

Wave shoaling

How does the wave height H vary?

- Normal wave incidence
- Wave energy conservation: **wave energy flux constant**

$$F_E = EC_g = c$$

$$C_g H^2 = c = C_{g0} H_0^2$$

with $E = \frac{\rho g H^2}{8}$

Shoaling coefficient

$$K_s = \frac{H}{H_0} = \sqrt{\frac{C_{g0}}{C_g}}$$

Wave shoaling

Shoaling coefficient:

$$K_s = \sqrt{\frac{n_0 C_0}{nC}}$$

$$n_0 = \frac{1}{2}$$

$$n = \frac{1}{2}\left(1 + \frac{2kh}{\sinh(2kh)}\right)$$

$$\frac{C}{C_0} = \tanh(kh)$$

$$K_s = \frac{1}{\sqrt{\tanh(kh)\left(1 + \frac{2kh}{\sinh(2kh)}\right)}}$$

$$K_s = \frac{\cosh(kh)}{\sqrt{kh + \sinh(kh)\cosh(kh)}}$$

Explicit function of kh but not of h

Wave shoaling

Proceed in 2 steps:

1. Show that kh varies like h:

$$\left.\frac{\partial kh}{\partial h}\right|_T = 2\pi \left.\frac{\partial (h/L)}{\partial h}\right|_T = \frac{2\pi}{L^2}\left(L - h\frac{\partial L}{\partial h}\right)$$

$$\frac{\partial L}{\partial h} = \frac{4\pi}{2kh + \sinh(2kh)}$$

$$\frac{\partial kh}{\partial h} = k\left(1 - \frac{2kh}{2kh + \sinh(2kh)}\right) > 0$$

Always positive

Therefore kh varies like h

Wave shoaling

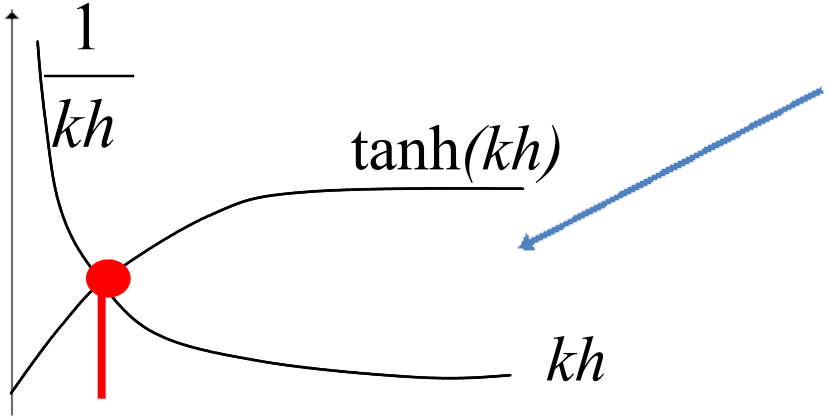
2. Study variations in kh by letting Y=kh:

$$\left.\frac{\partial Ks}{\partial Y}\right|_T = \frac{\partial}{\partial Y}\left[\frac{\cosh(Y)}{\sqrt{Y+\sinh(Y)\cosh(Y)}}\right] = \frac{\partial}{\partial Y}\left[\frac{\cosh(Y)}{\sqrt{A(Y)}}\right]$$

after a bit of algebra:

$$\left.\frac{\partial Ks}{\partial Y}\right|_T = A^{-3/2}\left[Y\sinh(Y) - \cosh(Y)\right]$$

derivative vanishes when $Y\tanh(Y)=1$ or $\tanh(kh)=\frac{1}{kh}$



single root: *kh* ≈1,198…

$\omega^2 = gk\tanh(kh) = gk_0$

$k_0h = 1$ or $\frac{h}{L_0} = \frac{1}{2\pi}$

Wave shoaling

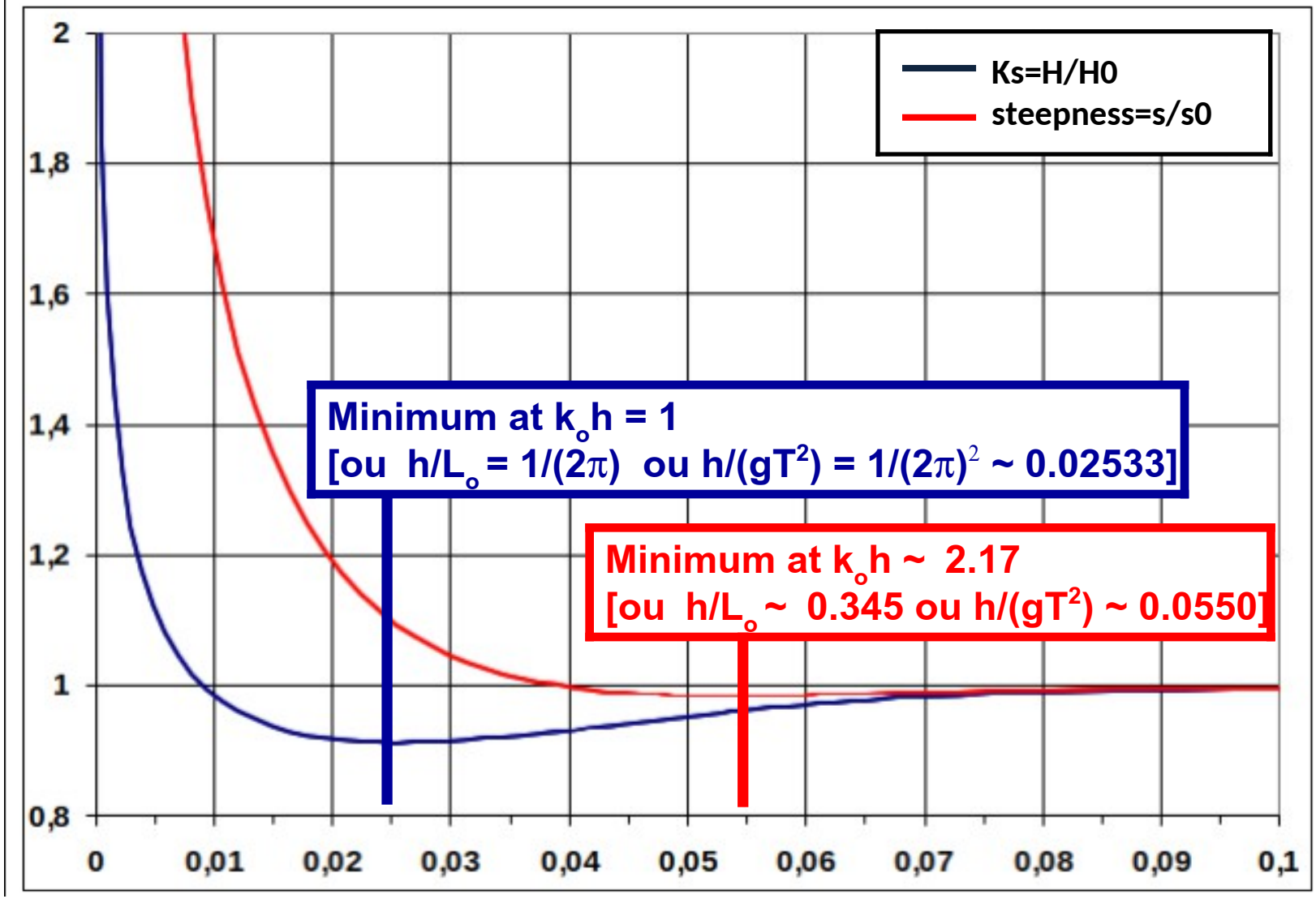
2. (cont) Thus Ks and H are NOT monotonic functions of h

As the water depth **h decreases**:
- the wave height **H decreases slightly** until $h/L_o = 1/(2\pi)$, with a **minimum wave height** $H = 0.913\, H_0$
- then the wave height **H increases** (theoretically, H-> ∞ as h -> 0, but in reality, waves break)

- the opposite is true for the **group velocity Cg**: increase to $h/L_o = 1/(2\pi)$ and then decrease to 0 as h->0

- similar behavior for the **wave steepness s=H/L**: decrease until $h/L_o = 0.345$, with a minimum of $s/s_0 = 0.985$, and then more rapid growth than *H* approaching the shore (*H* increases and L decreases)

Variations of Ks = H/H_o and s/s_o

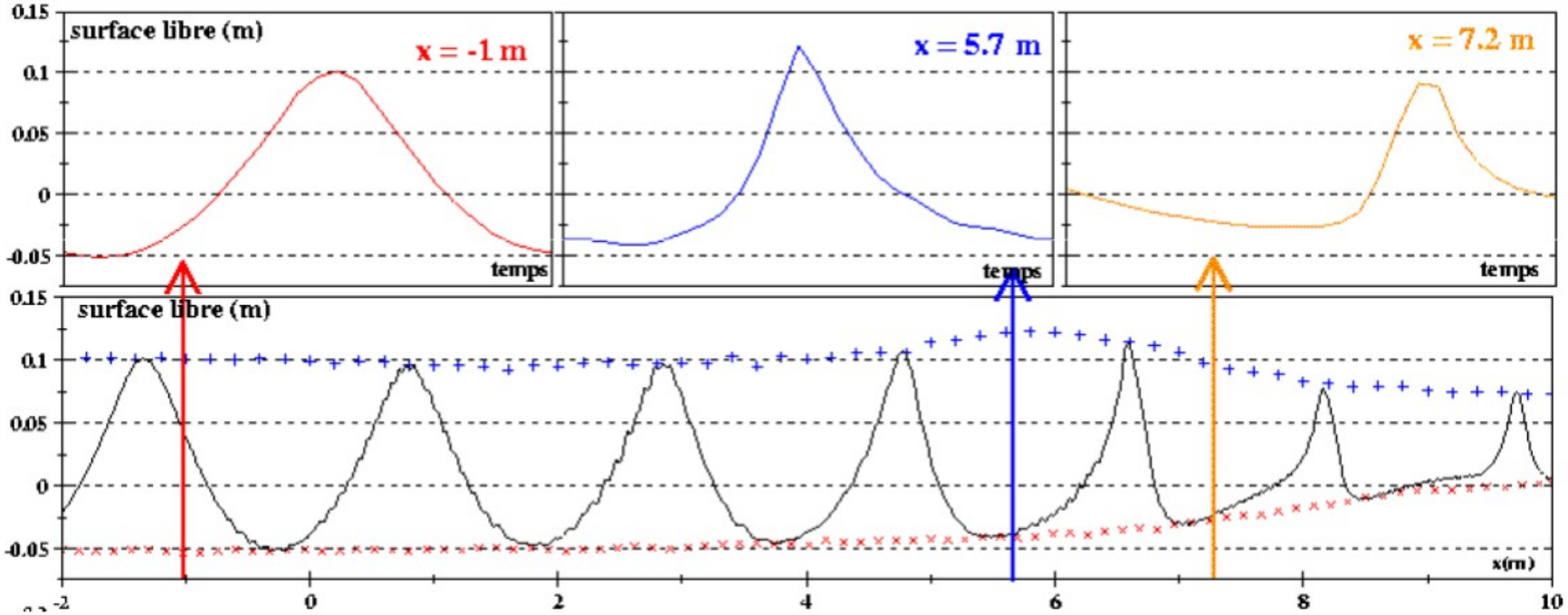


Effects of wave nonlinearity

- Linear waves:
  - Relative to the wavelength: $H/L \ll 1$
  - Relative to the water depth: $H/h \ll 1$

- Nonlinear waves:
  - What happens as these values increase?
  - To what maximum?
  - How does the wave shape change?

Wave nonlinearity



Sinusoidal waves:

Symmetric horizontally and vertically

Horizontal asymmetry:

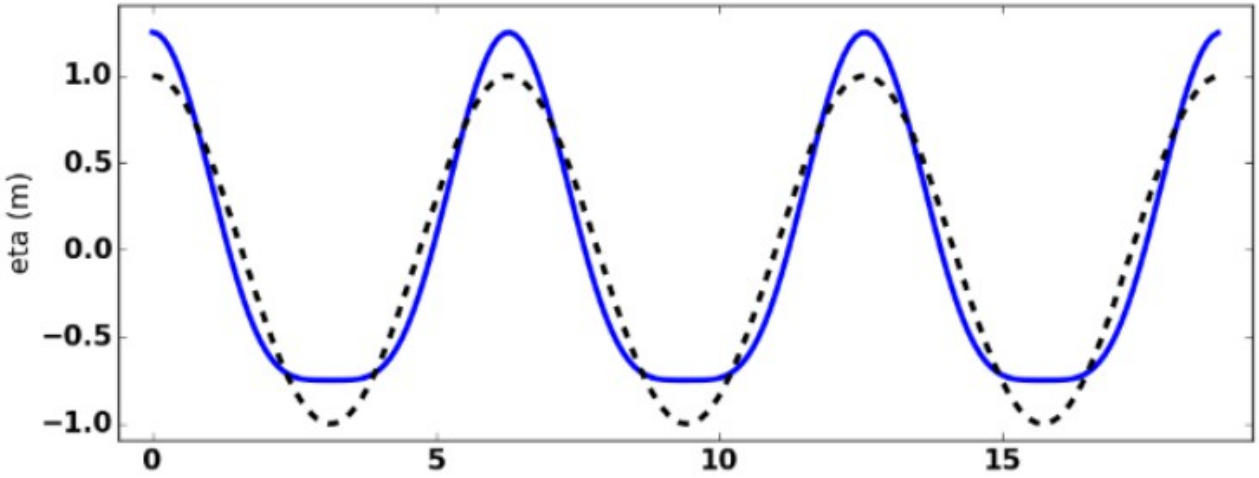
Wave crests are higher and narrower and wave troughs are shallower and wider

Vertical asymmetry:

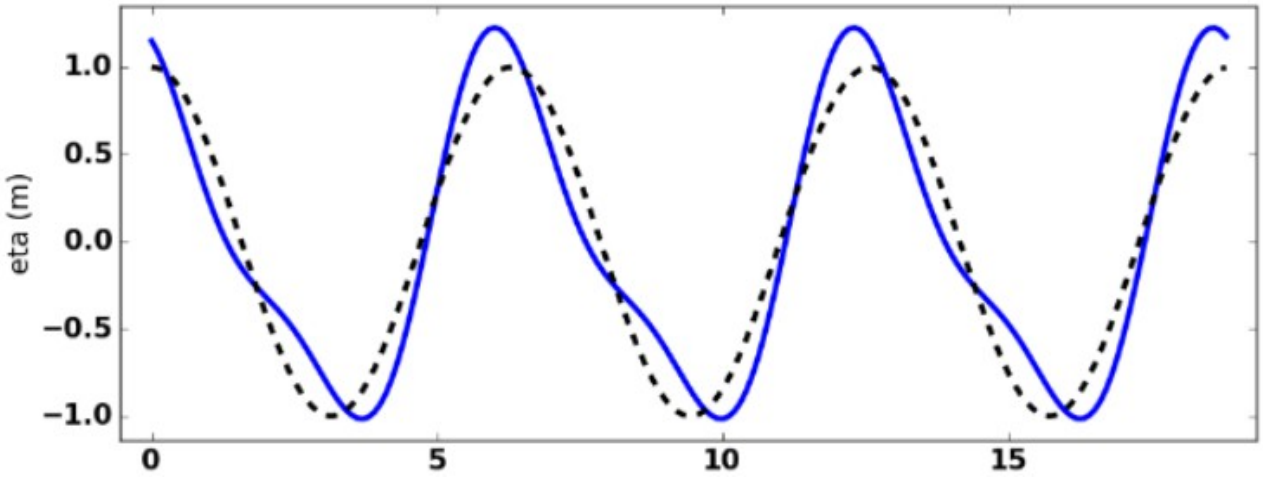
Wave fronts are steeper than the back

Nonlinéarités des vagues

Skewness (horizontal asymmetry)



Asymmetry (vertical asymmetry)



Review : Nondimensional numbers used to characterize waves

- **Dispersion: effects of *h***

  Do waves « feel » the bottom?

  Relative water depth:

  h/L or $\mu = kh$

- **Nonlinearity: effects of *H***

  Are waves a « large » perturbation of the free surface?

  Steepnees (deepwater):

  $s = H/L$ or $\epsilon = ka$

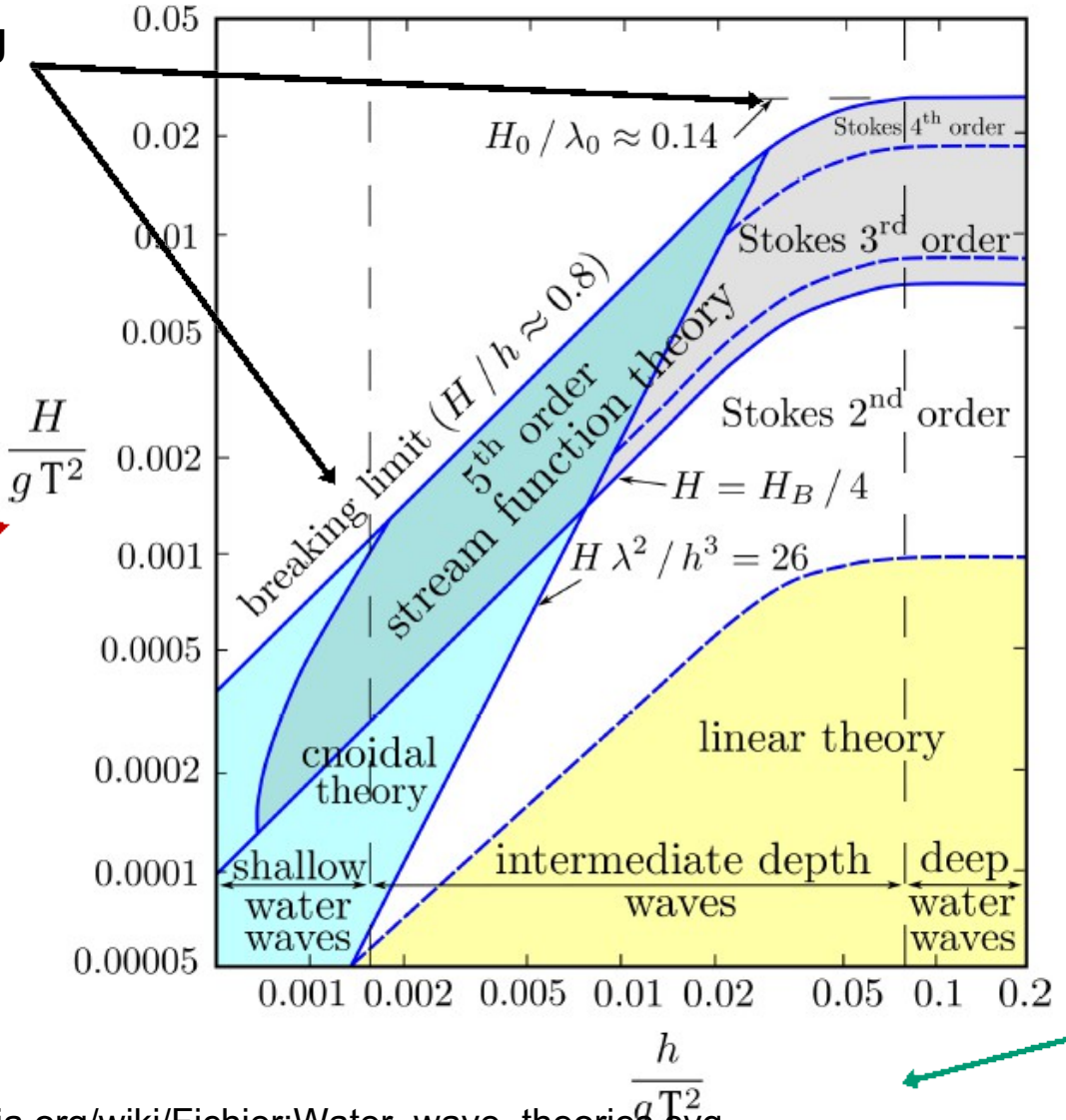
  Relative wave height (shallow water):

  H/h

- **Ursell number** = $U_s = \frac{HL^2}{h^3} = \left(\frac{H}{h}\right) \Big/ \left(\frac{h}{L}\right)^2$

Nondimensional numbers used to characterize waves

Wave breaking limits



Nonlinearity parameter

Dispersion parameter

From https://fr.m.wikipedia.org/wiki/Fichier:Water_wave_theories.svg

Nonlinear wave theories

- Flat bottom wave theories for periodic waves:

  Stokes, Cnoidal, stream function
- Nonlinear and dispersive wave models for variable bathymetry:
  Boussinesq , KdV, Serre, Green-Naghdi, …

Flat bottom wave theories for periodic waves : perturbation method

$$\phi = \varepsilon\phi^{(1)} + \varepsilon^2\phi^{(2)} + \ldots + \varepsilon^n\phi^{(n)}$$

$$\eta = \varepsilon\eta^{(1)} + \varepsilon^2\eta^{(2)} + \ldots + \varepsilon^n\eta^{(n)}$$

Stokes theory (deep water; high relative water depth) around the mean free surface, with **ε** = H/L

Cnoidal theory (shallow water; small relative water depth) around the bottom level, with **ε** = H/h

Stokes wave theory

Analytical solutions of the form:

$$\eta(x,t) = \frac{1}{k}\sum_n A_n \cos(n(kx - \omega t))$$

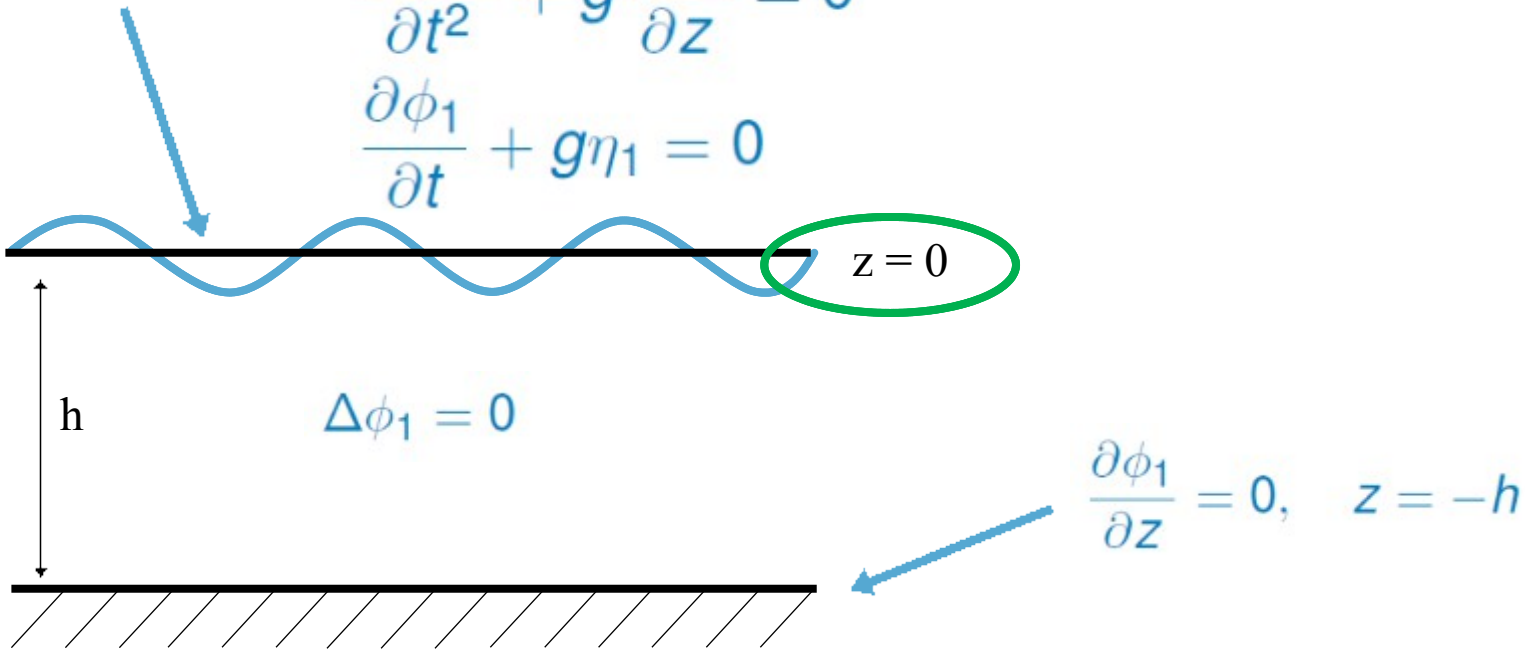
$$\phi(x,z,t) = \frac{c}{k}\sum_n B_n \cosh(nk(z+d))\sin(n(kx - \omega t))$$

Order of theory	**Authors / sources**
Order 1	Airy or Stokes (1847)
Order 2	Wiegel (1964)
Order 3	Skjelbreia (1958), Isobe & Kraus (1983)
Order 5	Skjelbreia & Hendrickson (1960), Isobe, Nishimura & Horikawa (1978), Fenton (1985)

Stokes wave theory: order 1

Order 1: $\frac{\partial^2 \phi_1}{\partial t^2} + g\frac{\partial \phi_1}{\partial z} = 0$

$\frac{\partial \phi_1}{\partial t} + g\eta_1 = 0$



Airy wave theory!

$$\eta = a\cos(kx - \omega t)$$

$$\phi = \frac{ag}{\omega}\frac{\cosh(k(z+h))}{\cosh(kh)}\sin(kx - \omega t)$$

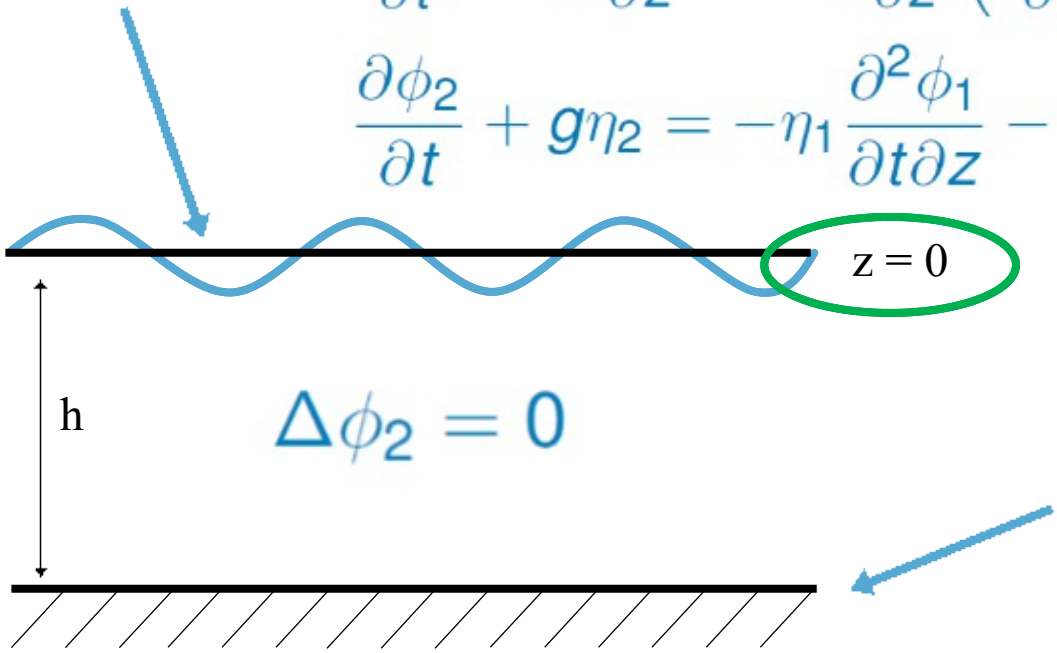
$$\omega^2 = gk\tanh(kh)$$

Stokes wave theory: order 2

Order 2:

$$\frac{\partial^2\phi_2}{\partial t^2} + g\frac{\partial\phi_2}{\partial z} = -\eta_1\frac{\partial}{\partial z}\left(\frac{\partial^2\phi_1}{\partial t^2} + g\frac{\partial\phi_1}{\partial z}\right) - \frac{\partial}{\partial t}(|u_1|^2)$$

$$\frac{\partial\phi_2}{\partial t} + g\eta_2 = -\eta_1\frac{\partial^2\phi_1}{\partial t\partial z} - \frac{1}{2}|u_1|^2$$



$$\Delta\phi_2 = 0$$

$$\frac{\partial\phi_2}{\partial z} = 0, \quad z = -h$$

Uses first order Stokes solution in RHS → expressions now contain second-order correction terms

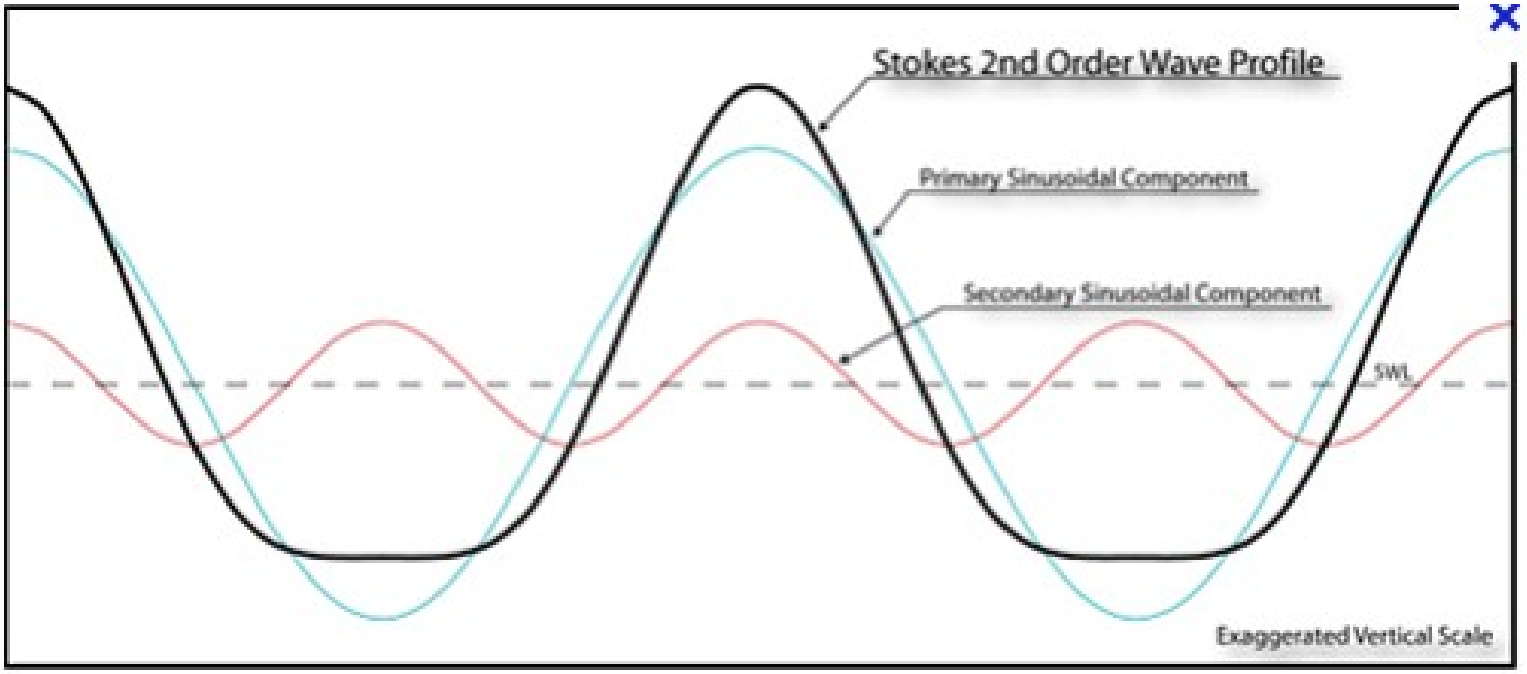
Stokes order 2 (Wiegel, 1964)

Dispersion relation: $\omega^2 = gk\tanh(kh)$ or $L = L_0\tanh(kh)$, $L_0 = \frac{gT^2}{2\pi}$

(same as order 1: only changes for ODD orders)

Free surface: $\eta(x,t) = \underbrace{\frac{H}{2}\cos\theta}_{\text{Order 1}} + \underbrace{\frac{kH^2}{16}\frac{\cosh(kh)}{\sinh^3(kh)}(2+\cosh(2kh))\cos(2\theta)}_{\text{Order 2}}$

$\theta = kx - \omega t$



Stokes order 2 (Wiegel, 1964)

Then calculate orbital velocities, pressure, potential…

Orbital velocities:

$$u(x,t) = \frac{\omega H}{2}\frac{\cosh(\,k(z+d))}{\sinh(\,kd\,)}\cos\,\theta + \frac{3}{4}\frac{\omega H}{2}\frac{kH}{2}\frac{\cosh(\,2k(z+d))}{\sinh^{\,4}(kd\,)}\cos(\,2\theta)$$

$$w(x,t) = \frac{\omega H}{2}\frac{\sinh(\,k(z+d))}{\sinh(\,kd\,)}\sin\,\theta + \frac{3}{4}\frac{\omega H}{2}\frac{kH}{2}\frac{\sinh(\,2k(z+d))}{\sinh^{\,4}(kd\,)}\sin(\,2\theta)$$

Pressure:

$$p(x,t) = -\rho gz + \frac{1}{2}\rho gH\,\frac{\cosh(\,k(z+d))}{\cosh(\,kd\,)}\cos\,\theta$$

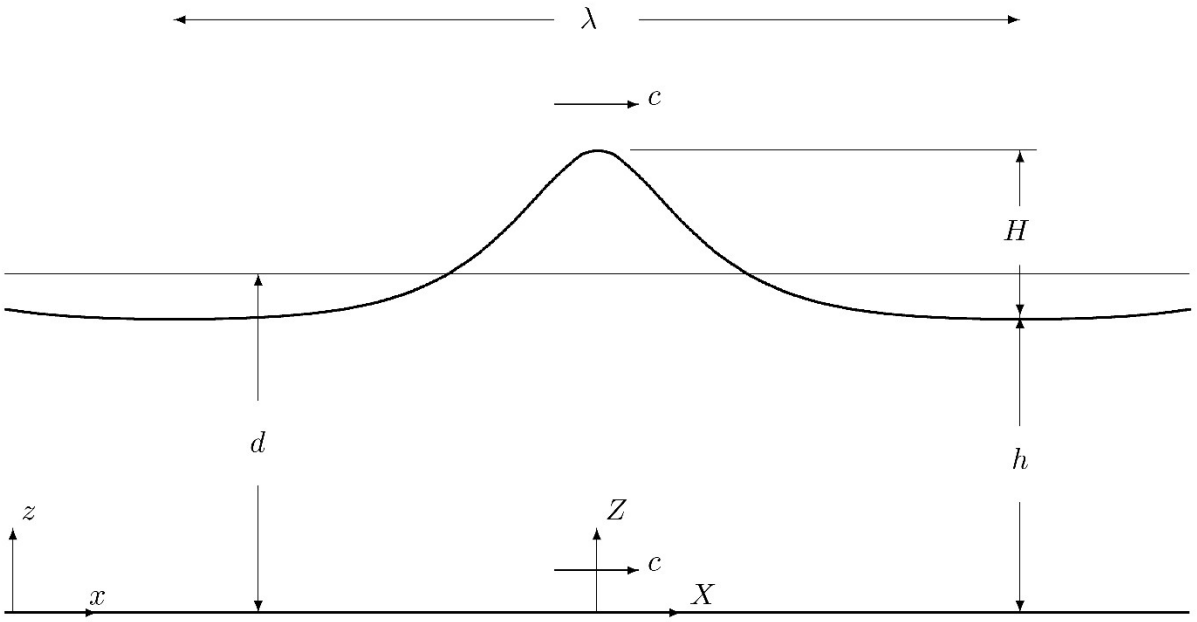
$$+\frac{3}{4}\rho gH\left(\frac{\pi H}{L}\right)\frac{1}{\sinh(\,2kd\,)}\left(\frac{\cosh(\,2k(z+d))}{\sinh^{\,2}(kd\,)} - \frac{1}{3}\right)\cos(\,2\theta)$$

$$-\frac{1}{4}\rho gH\left(\frac{\pi H}{L}\right)\frac{1}{\sinh(\,2kd\,)}(\cosh(\,2k(z+d)) - 1)$$

Then on to order 3……

Cnoidal wave theory

- Wave of constant form in steadily moving frame of reference
- Velocity expressed as a function of the stream function
- 2 ODE's of infinite order, must be approximated:
  - Series method (traditional approach)
  - Numerical spectral approach (similar to stream function theory)



Fenton (1998)

Cnoidal wave theory

- Analytical solutions using elliptical functions cn, dn, sn
- Example at order 1 (Korteweg and de Vries, 1895; Isobe, 1985):

$$\eta(x,t) = H\left(cn^2 - \overline{cn^2}\right)$$

$$u(x,t) = \left(\frac{g}{h}\right)^{1/2} \eta(x,t)$$

$$w(x,t) = \left(\frac{g}{h}\right)^{1/2} H \frac{4Kh}{L} \frac{z+h}{h} cn(\theta,m).sn(\theta,m).dn(\theta,m)$$

$$L = T\sqrt{gh} \qquad \frac{gHT^2}{2\pi} = \frac{16}{3} m^2 K^2 \qquad K = \int_0^{\pi/2} \frac{d\Phi}{\sqrt{1 - m^2 \sin^2 \Phi}}$$

Stream function theory

- Moving reference frame with celerity C
- Introduce stream function ψ(x, z, t):

$$u(x,z,t) = \frac{\partial \psi}{\partial z}; w(x,z,t) = -\frac{\partial \psi}{\partial x}$$

- Fourier series solution of the form:

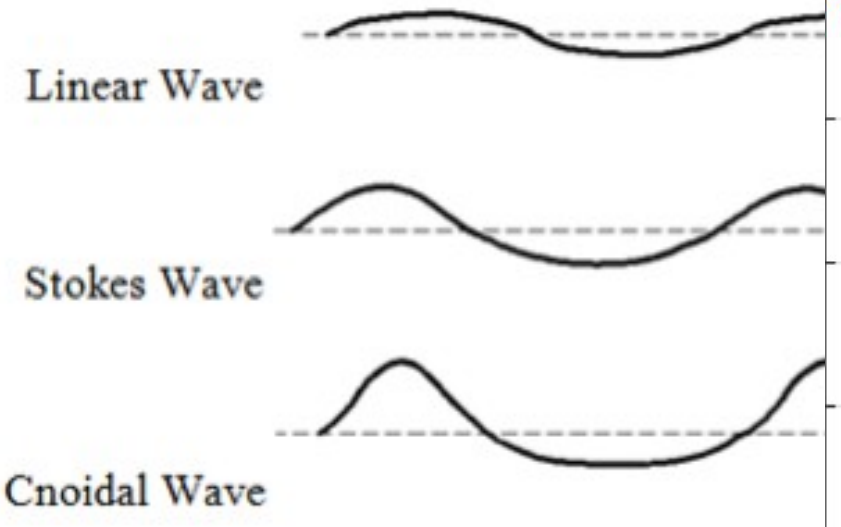
$$\psi(x,z) = -\bar{u}(z+h) + \sqrt{\frac{g}{k^3}} \sum_{j=1}^{N} B_j \frac{\sinh(jk(z+h))}{\cosh(jkh)} \cos(jkx)$$

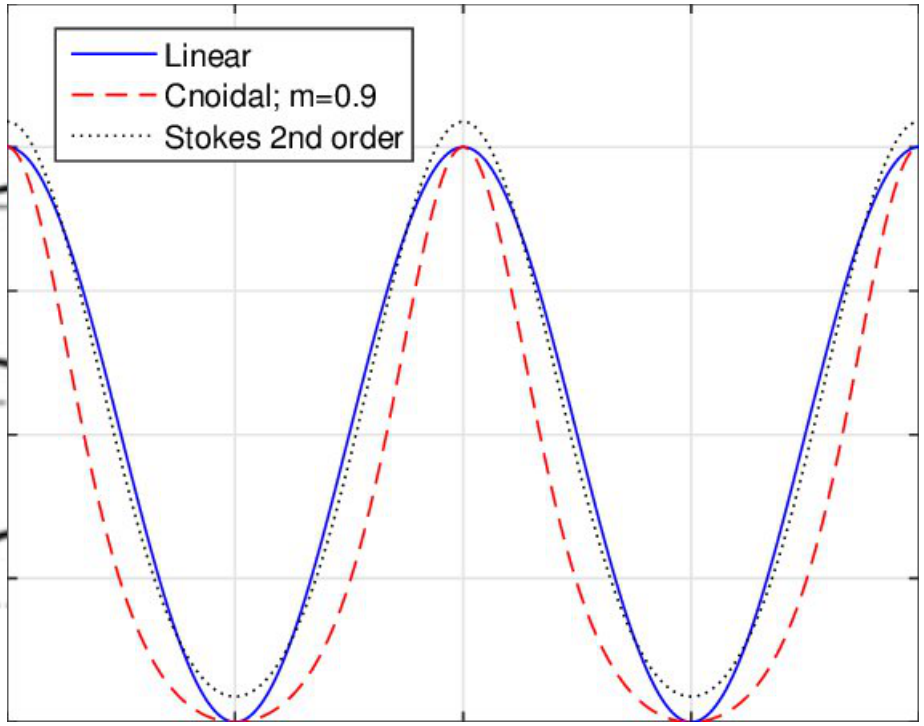
  - Solutions to the Laplace equation with periodic boundary conditions + impermeable bottom,
  - Solve the fully nonlinear and dispersive kinematic and dynamic free surface boundary conditions

Summary of flat bottom

Stokes theory (deep water; large relative water depth) around the mean free surface, with **ε** = H/L

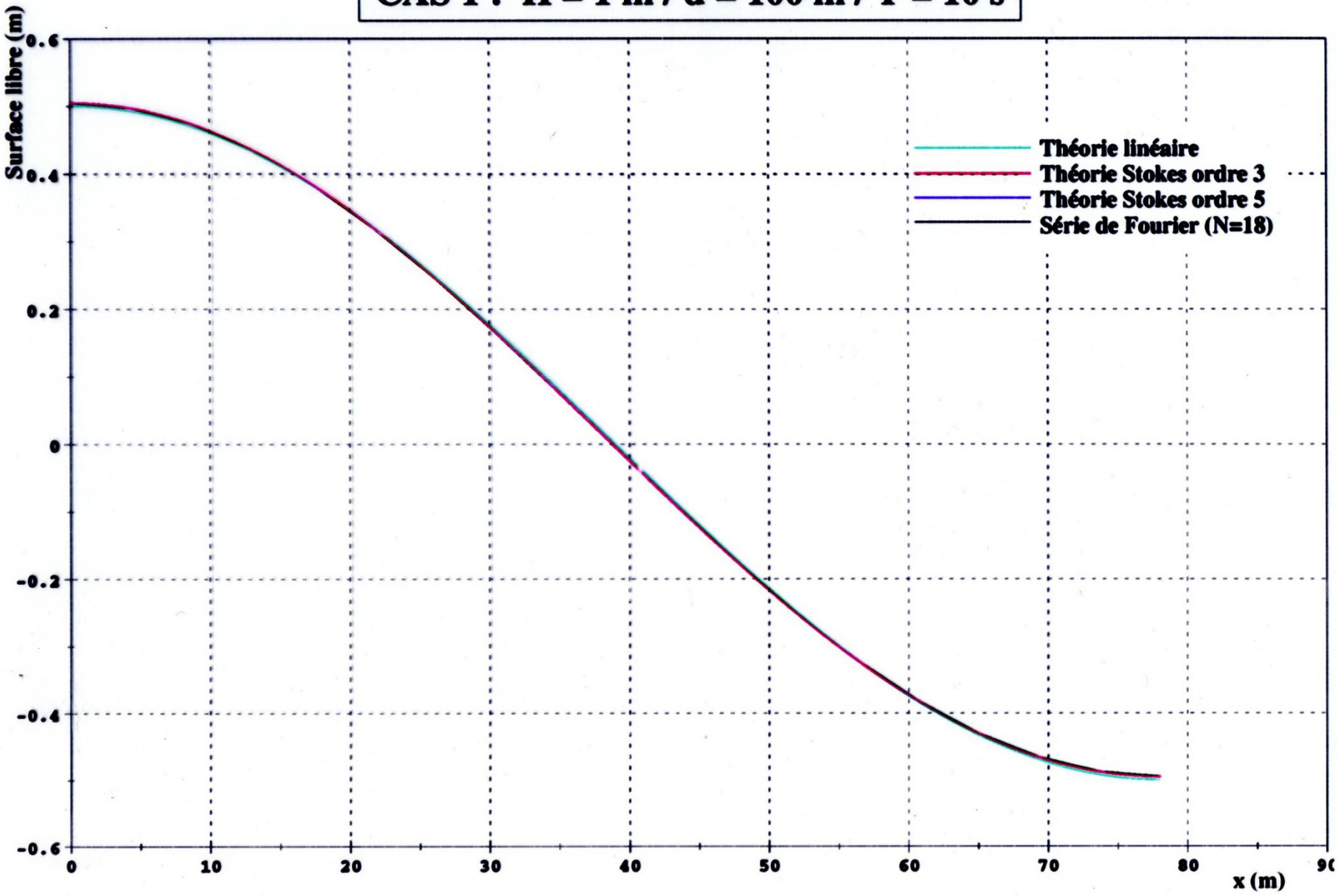
Cnoidal theory (shallow water; small relative water depth) around the bottom level, with **ε** = H/h



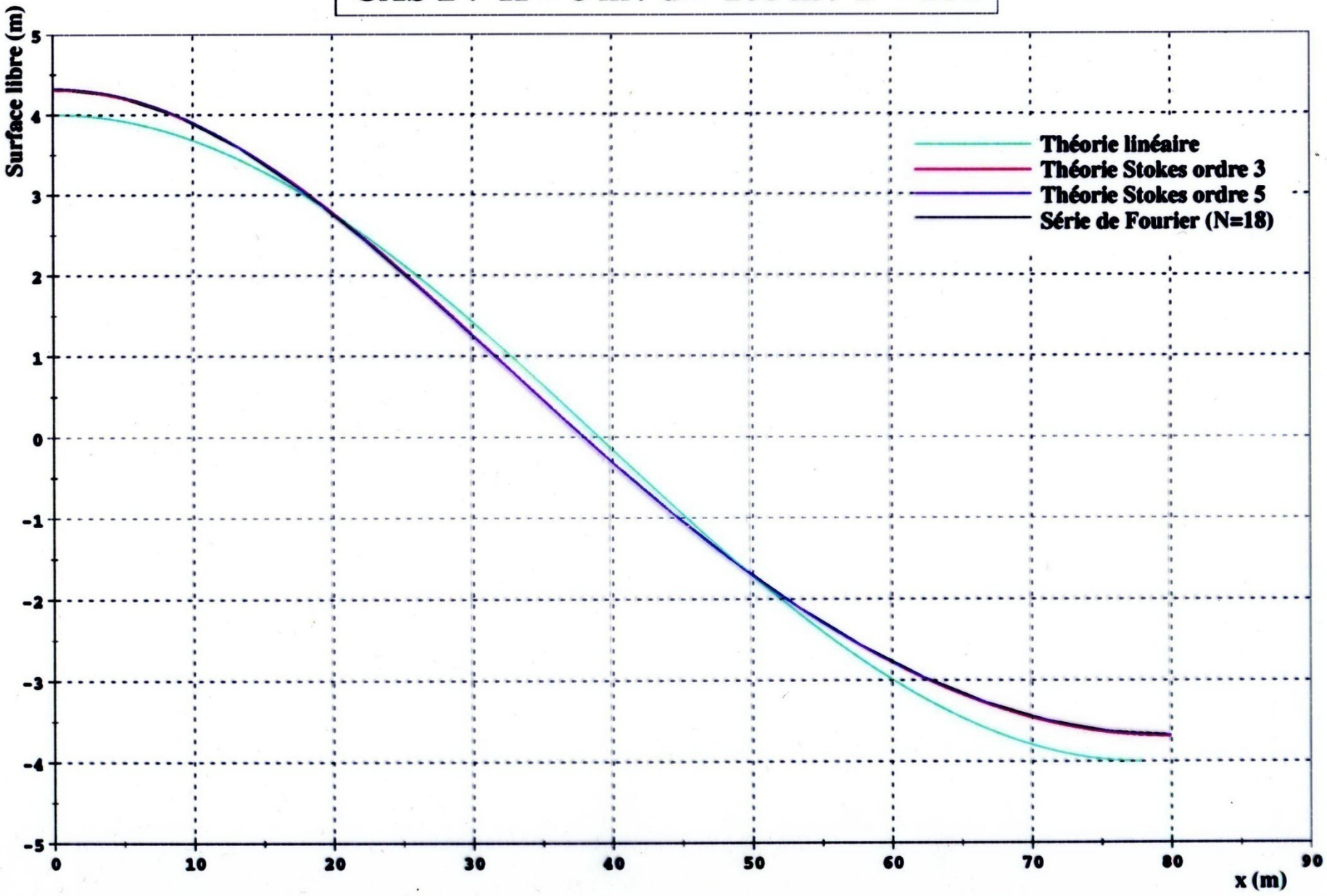


https://www.researchgate.net/publication/318588076_Experimental_study_of_water_waves_nonlinear_effects_and_absorption/figures?lo=1

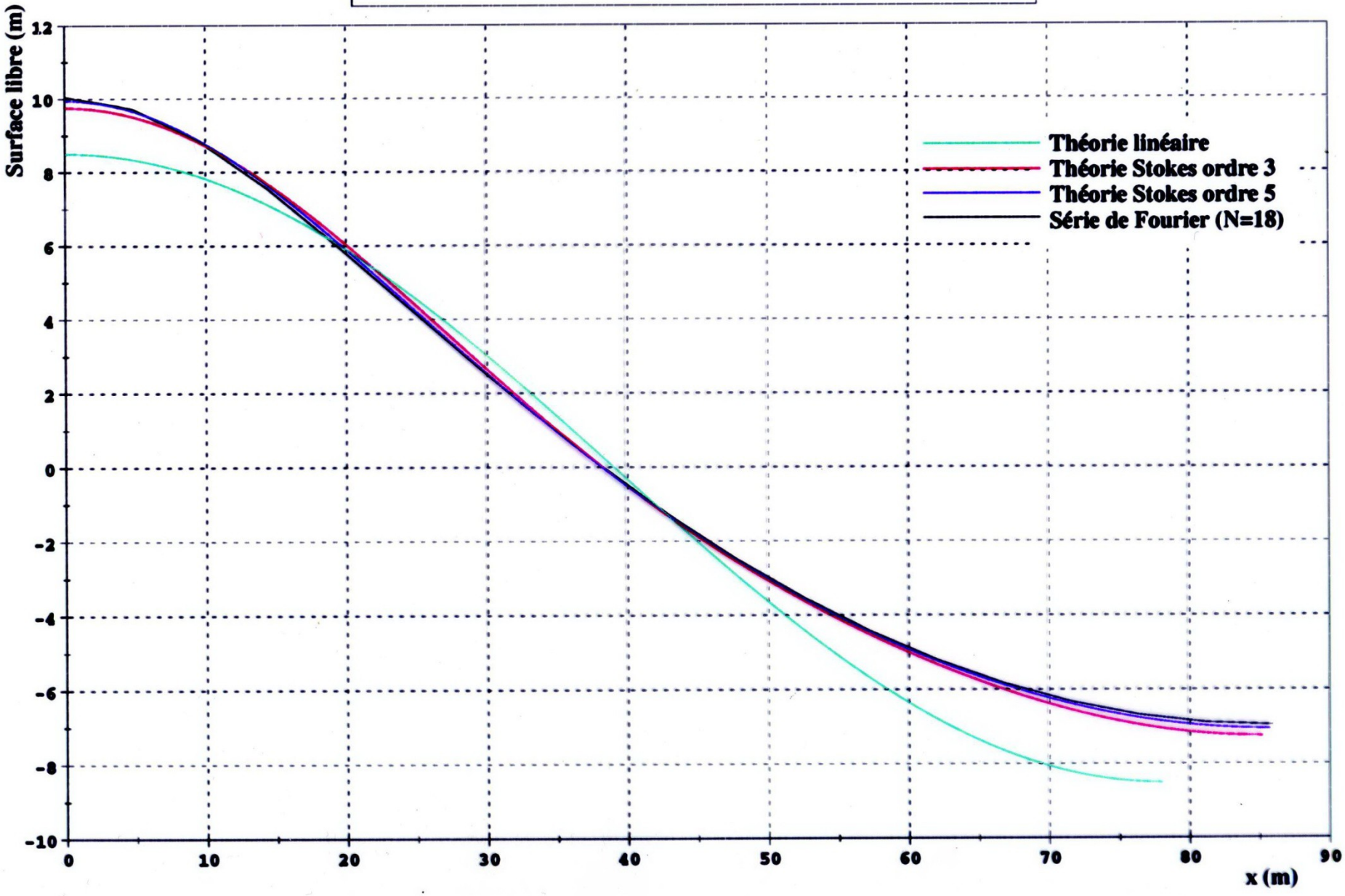
CAS 1 : H = 1 m / d = 100 m / T = 10 s



CAS 2 : H = 8 m / d = 100 m / T = 10 s



CAS 3 : H = 17 m / d = 100 m / T = 10 s



Nonlinear wave theories

	Shallow water theory	Transition zone	Deep water theory
Nonlinear wave theory	**Cnoidal theory**		**Stokes theory**
	← Stream function theory →		
Relative water depth	$h/L < 1/25$	$1/25 < h/L < 1/2$	$h/L > 1/2$
Nondimensional period	$T(g/h)^{1/2} > 25$	$4 < T(g/h)^{1/2} < 25$	$T(g/h)^{1/2} < 4$
Ursell number	$U_s > 40$	$10 < U_s < 40$	$U_s < 10$

Course schedule

I. Wave dynamics	Project	Lecturer	Date
1. Introduction and characterizing ocean waves and sea states	Description of class project and site selection	M. Yates	25/09
2. Linear wave theory	Characterizing offshore waves	M. Yates	09/10
3. Nearshore wave propagation	**Calculating extreme wave conditions**	**M. Yates**	**16/10**
4. Coastal hydrodynamics	**Student mini-presentations**	M. Yates	**06/11****
II. Numerical wave modeling			
5. Numerical wave modeling I	Introduction to wave propagation model	J. Harris	**23/10****
6. Numerical wave modeling II	Estimating mean and extreme nearshore waves	J. Harris	13/11
III. Wave-structure interactions			
7. Dynamics of a body in waves	Wave load estimation	J. Harris	20/11
8. Modeling wave-body interactions	Calculating movement of floating wind turbine	C. Peyrard	04/12
9. Offshore wind turbine foundations	Final project work session	C. Peyrard	11/12
10. Final project presentations 14h - 18h15			***18/12***